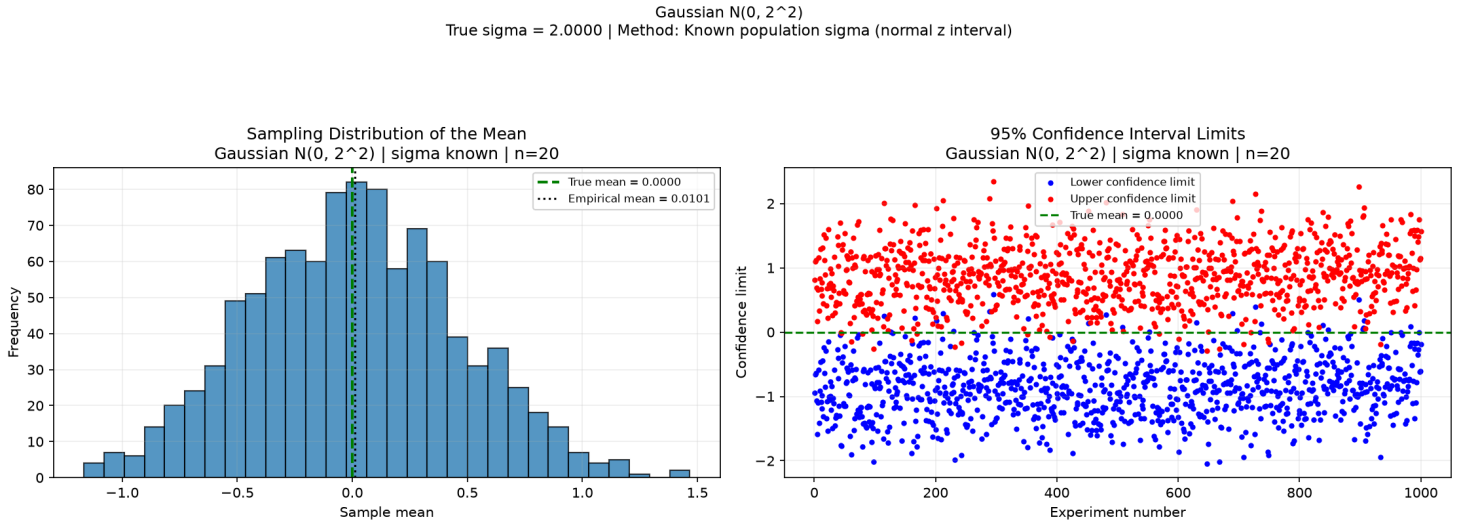
Homework_3_solutions

1.Write a program to do the following:
a) Generate n i.i.d random variables of a specified distribution F whose mean and standard deviation are known.
(b) Compute the sample mean and the limits of the 95% confidence interval.
(c) Check if the true mean is in the 95% confidence interval;
(d) In an outer loop, repeat this experiment K times.
(e) Plot the histogram of the sample mean from the K experiments.
(f) Represent the lower limit by a blue dot and the right limit by a red dot and make a scatter plot of both these limits.

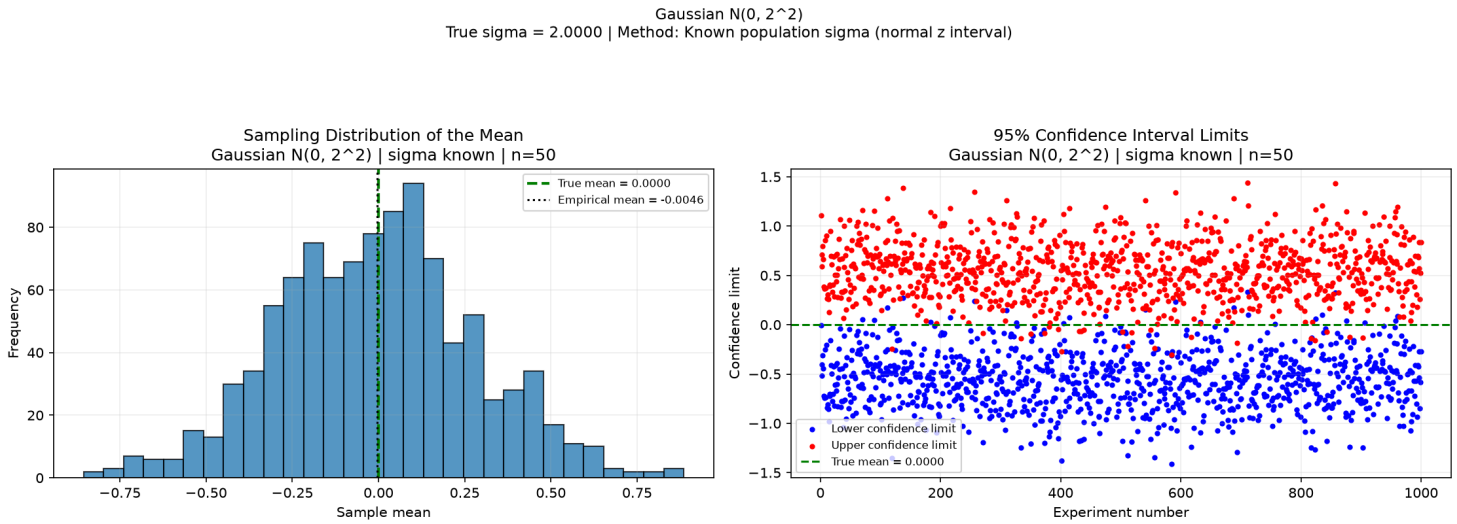
Once you have understood what you need to do, we will just change F, n, and K and see how things behave. This is next

(a) Let F correspond to Gaussian with μ = 0 and σ = 2. Let n = 20 and K = 1000. Repeat for different σ = 3,5 and n = 50,100,200,1000. For each combination of σ and n, print the histogram and the scatter plot. Also mention the number of times that the true mean was inside the confidence interval calculate.

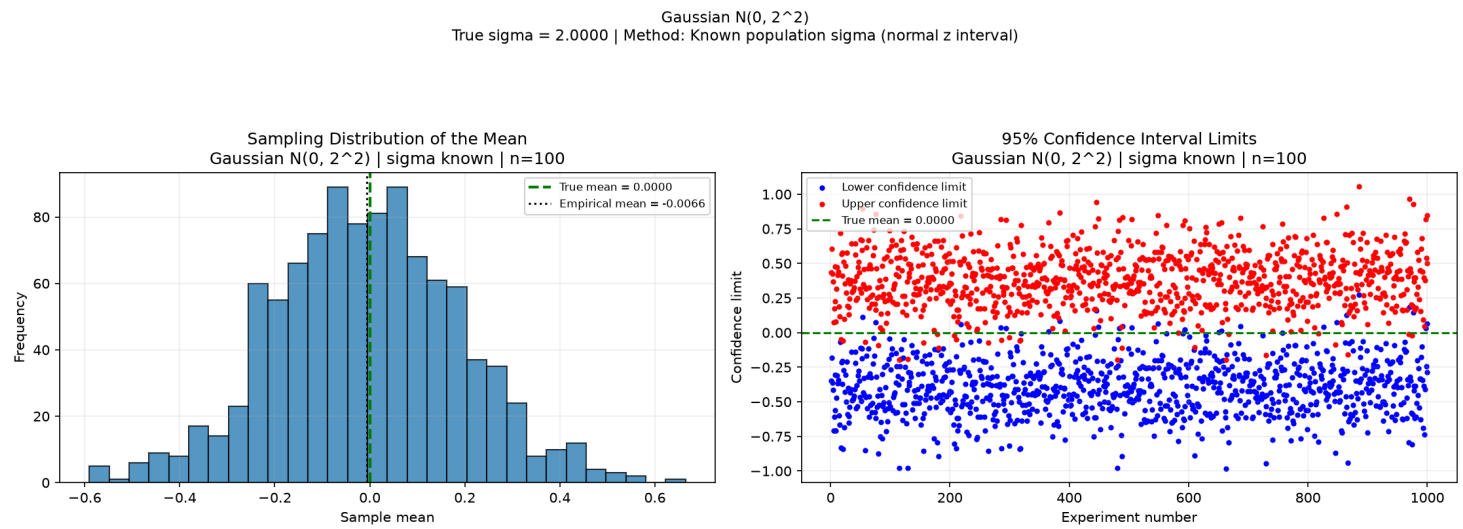
σ = 2 and n = 20



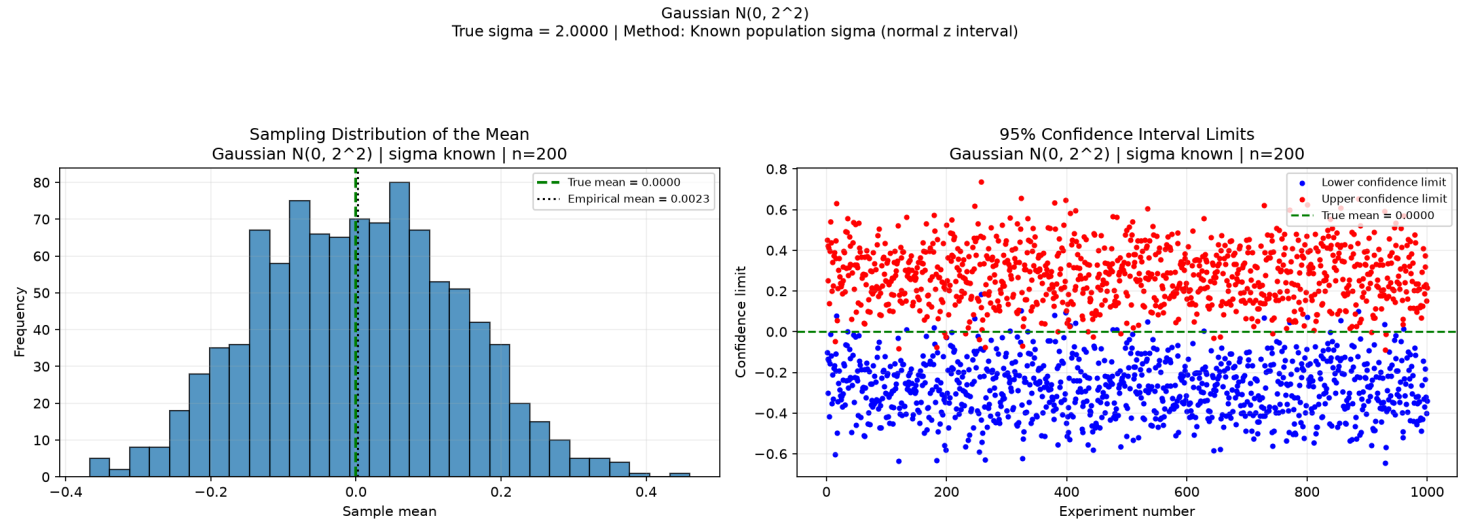
σ = 2 and n = 50



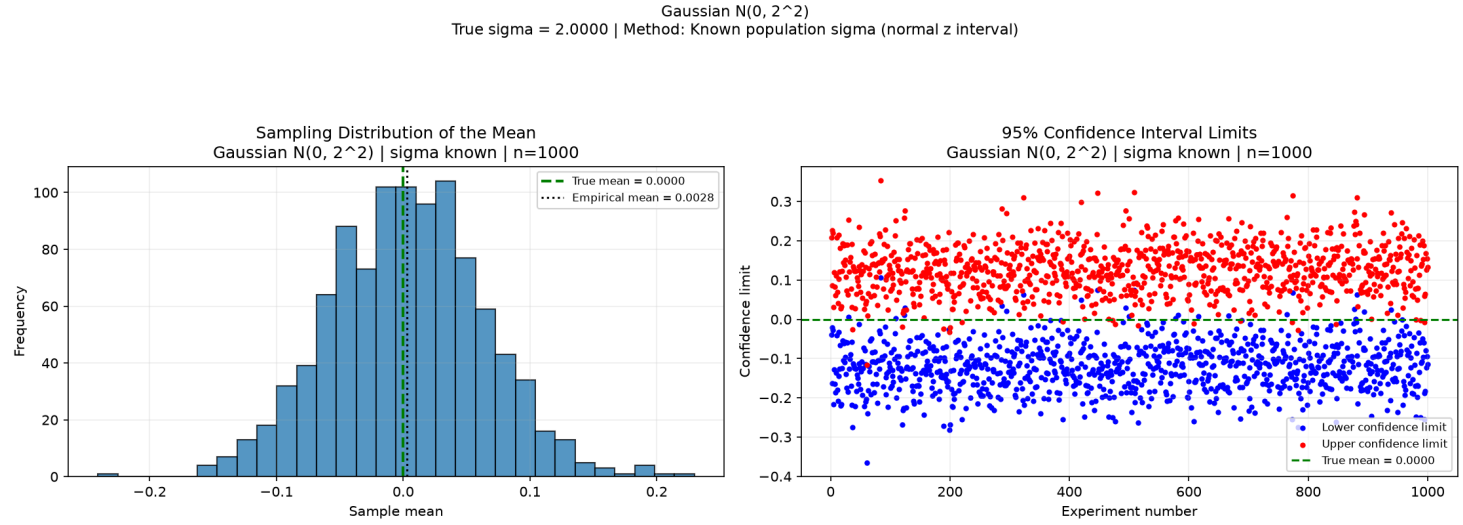
σ = 2 and n = 100



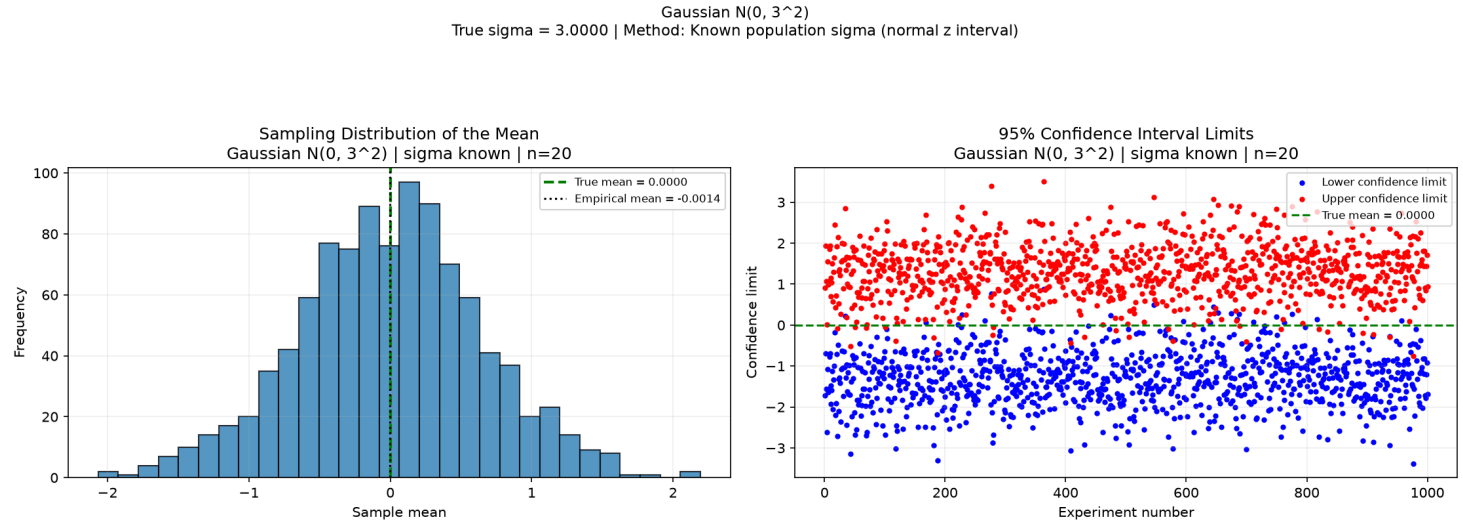
σ = 2 and n = 200



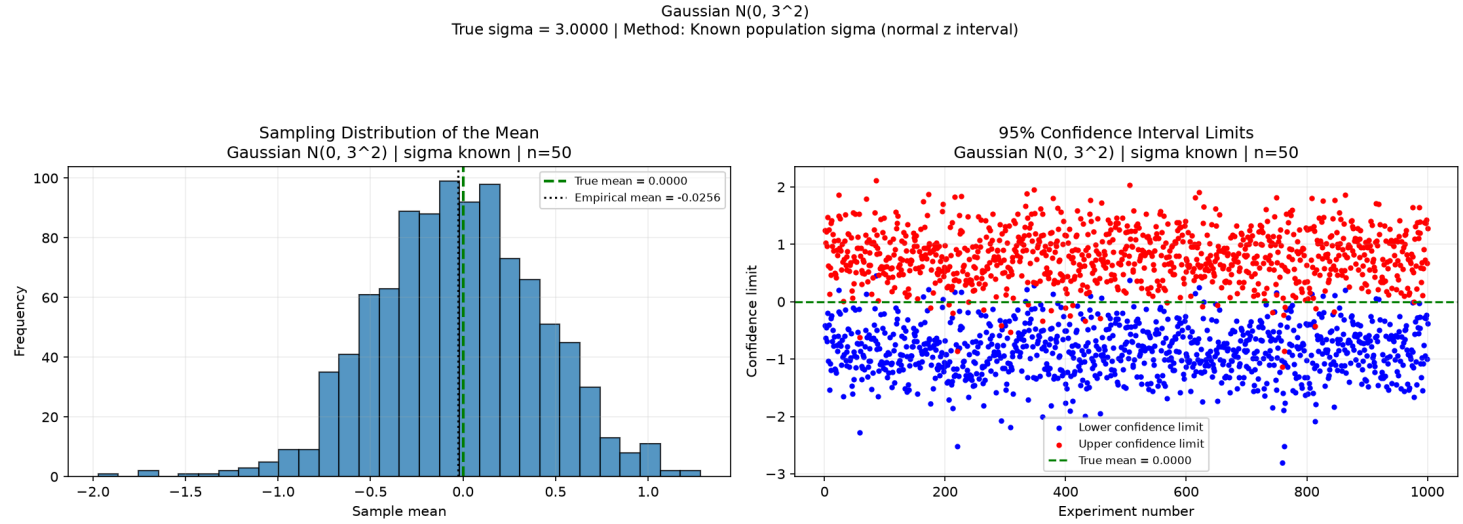
σ = 2 and n = 1000



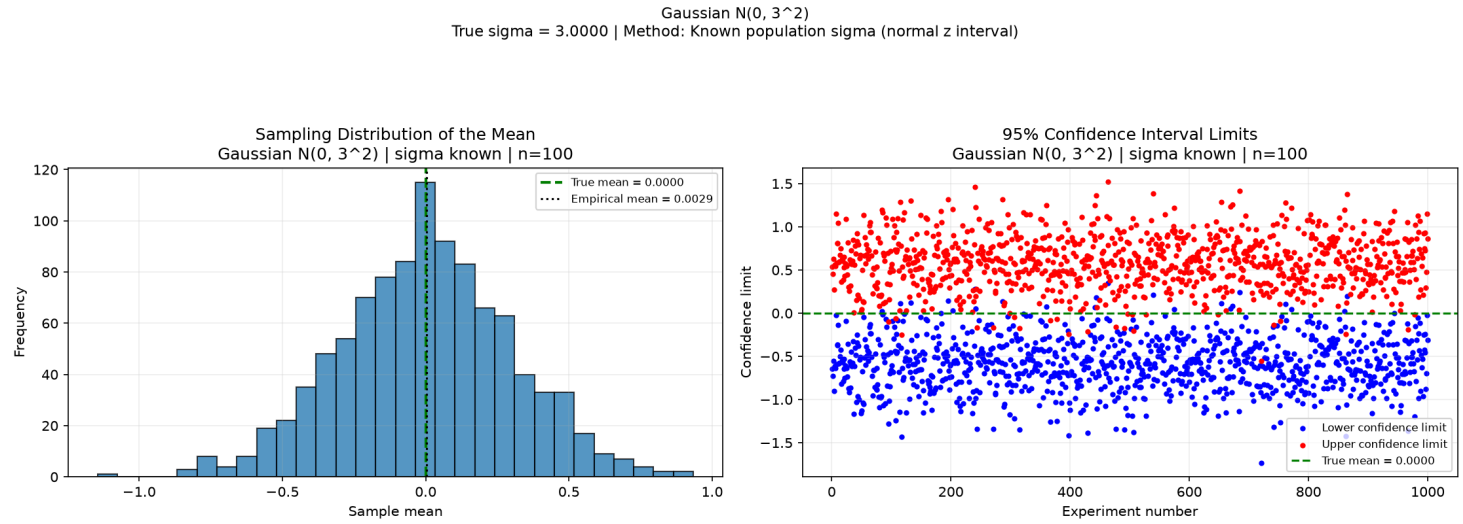
σ = 3 and n = 20



σ = 3 and n = 50

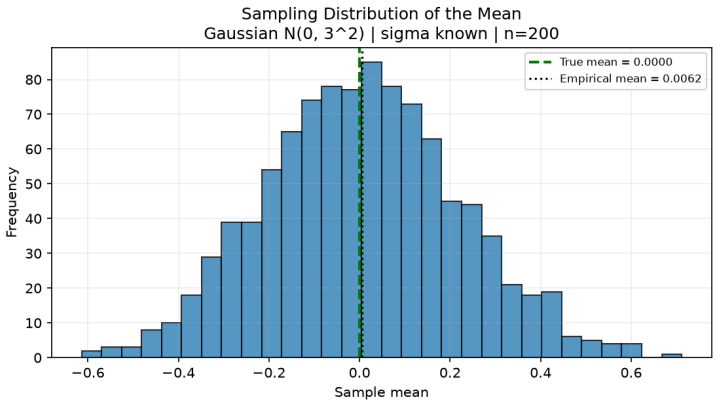


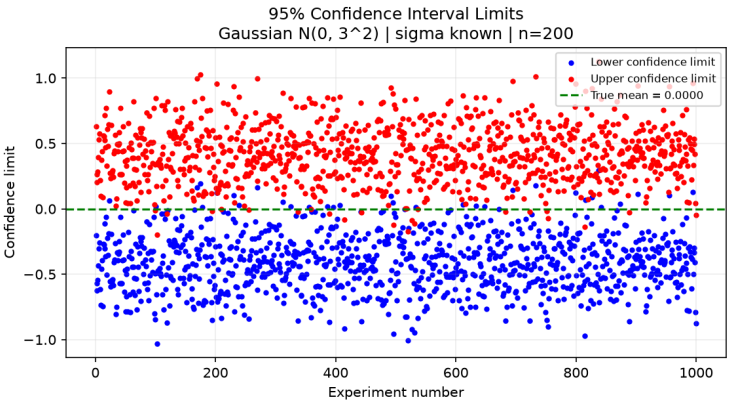
σ = 3 and n = 100



σ = 3 and n = 200

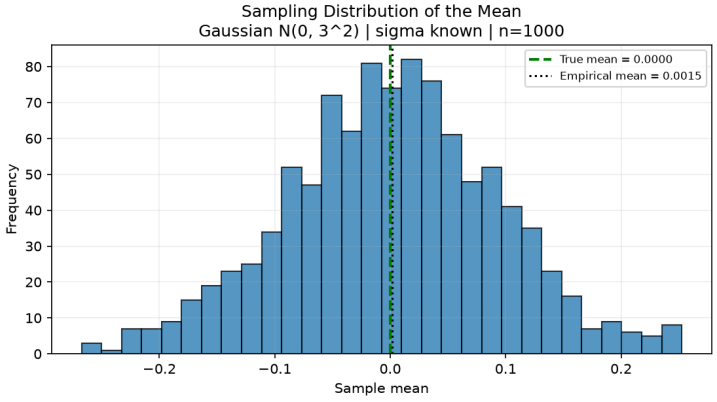
Gaussian N(0, 3^2)
True sigma = 3.0000 | Method: Known population sigma (normal z interval)

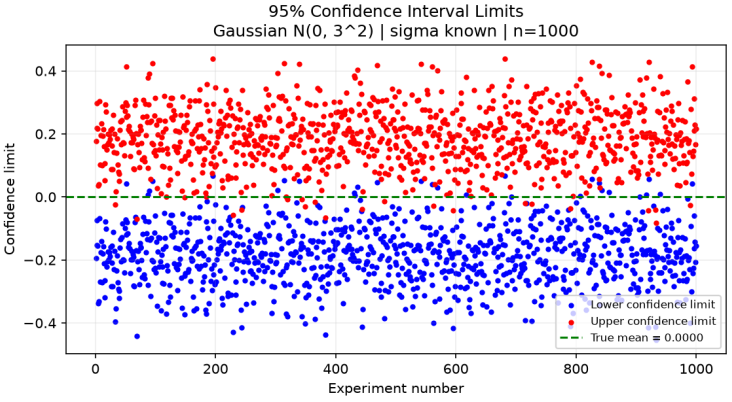




σ = 3 and n = 1000

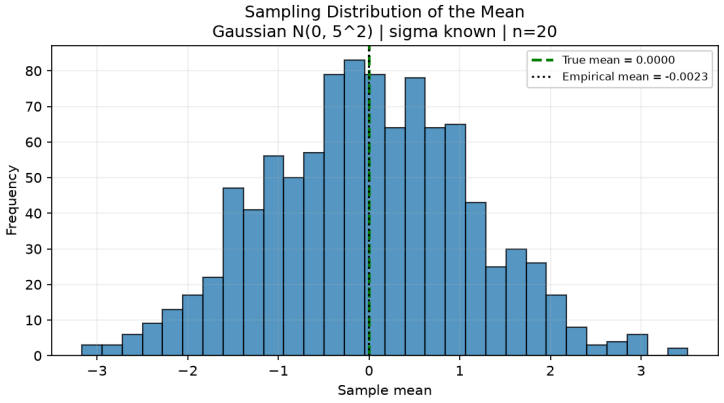
Gaussian N(0, 3^2)
True sigma = 3.0000 | Method: Known population sigma (normal z interval)

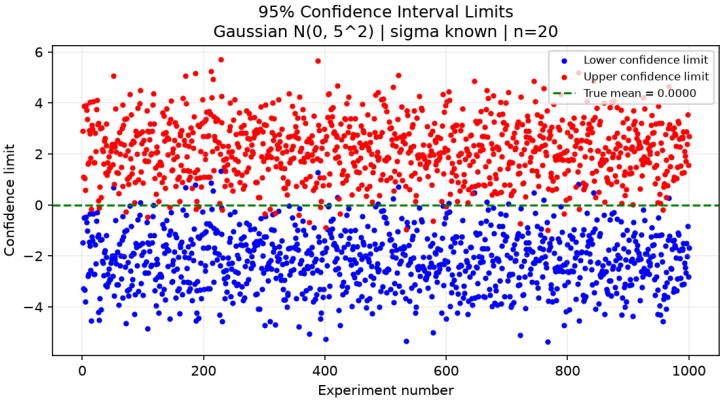




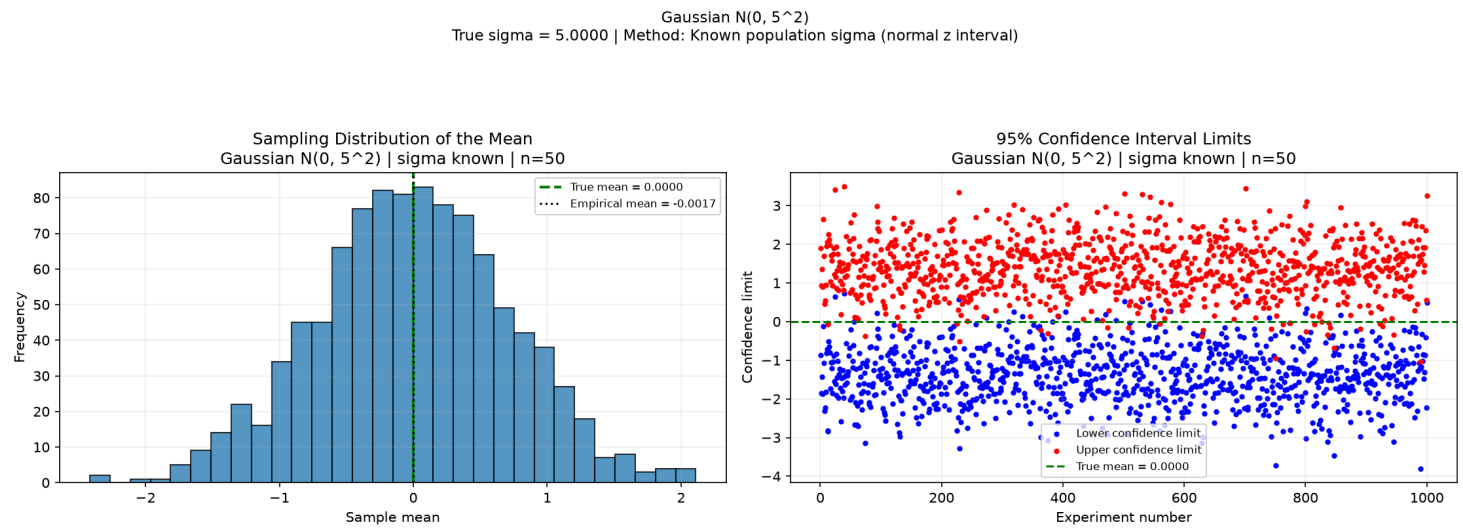
σ = 5 and n = 20

Gaussian N(0, 5^2)
True sigma = 5.0000 | Method: Known population sigma (normal z interval)

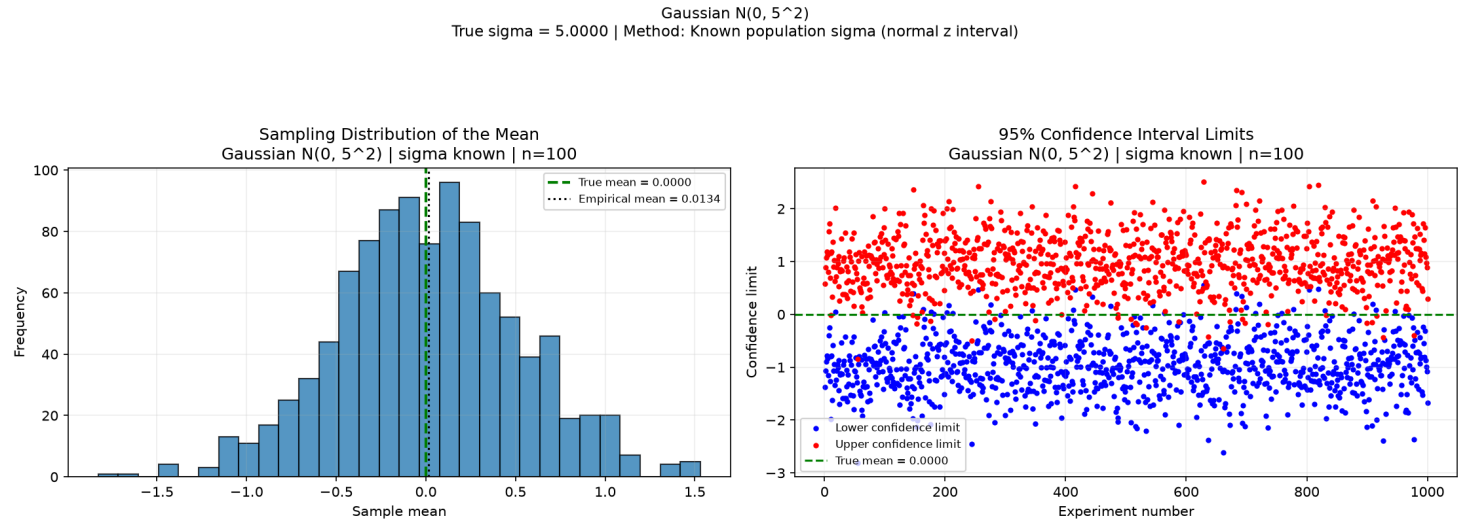




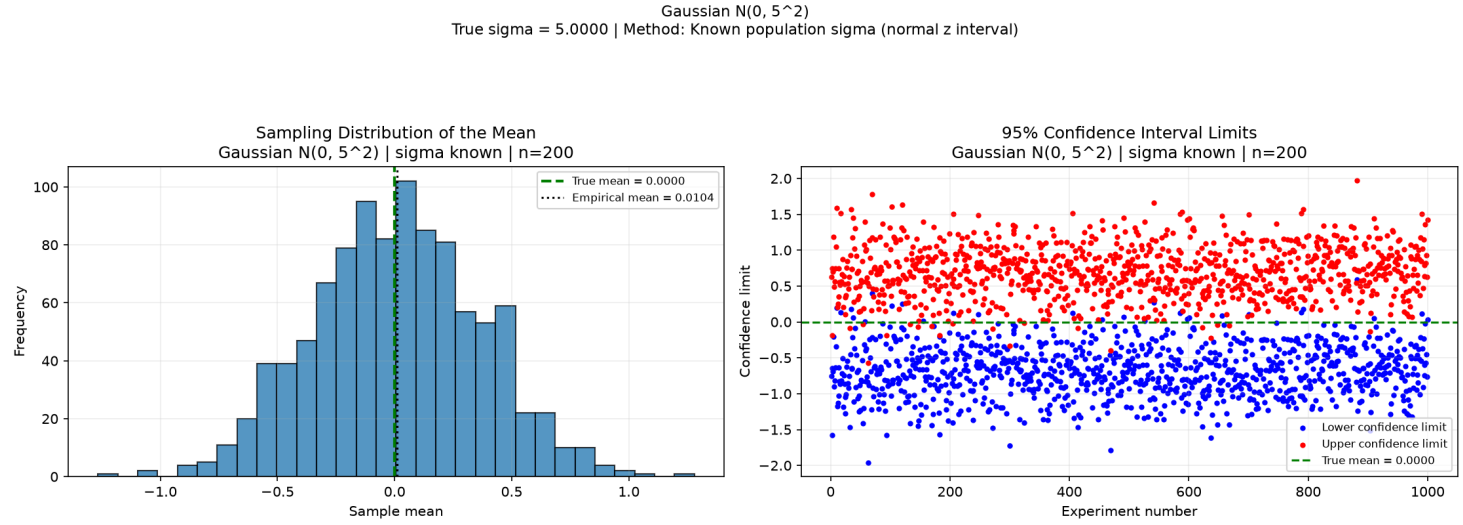
σ = 5 and n = 50



σ = 5 and n = 100

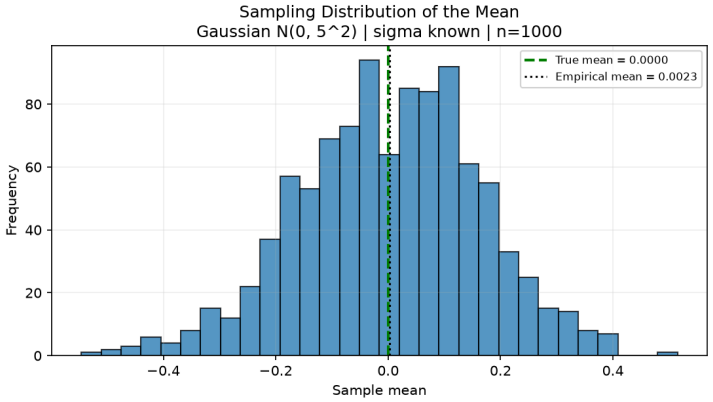


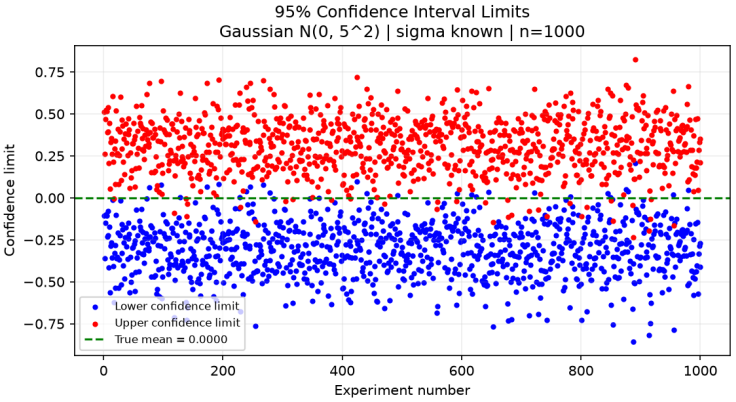
σ = 5 and n = 200



σ = 5 and n = 1000

Gaussian N(0, 5^2)
True sigma = 5.0000 | Method: Known population sigma (normal z interval)





Statistical Foundations of Machine Learning

Homework 3

1)

(a) Gaussian with known sigma

Underlying experiment :- Consider a population with mean = 0 & known standard deviation $\sigma = 2$

$$X \sim N(0, 2^2)$$

Of the population we first select n = 20 samples which are supposed to be independent & identically distributed. Now, we calculate the sample mean. We then repeat this exercise k times, which is 1000 in our case. Each sample has its own sample mean. Based on the question we have conducted this experiment for $\sigma = 2, 3, 5$ & $n = 20, 50, 100, 200, 1000$. For each experiment we have the histogram of the sample means & a scatter plot of the confidence-interval limits. Before we can get into the inference from these plots, let us look at some calculations:-

Consider $n = 20, \mu = 0, \sigma = 2$

Suppose the sample mean in this case is 0.4.

→ Standard error

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{20}} \approx 0.4472$$

Confidence interval = Point estimate ± Margin of error

= Point estimate ± (Critical value × Standard error)

= 0.4 ± (1.96 × 0.4472)

= 0.4 ± 0.8765

which gives [-0.4765, 1.2765]

The true population mean, $\mu = 0$ falls in this confidence interval. This is the calculation for 1 of the 1000 runs (k-value). If 950 of these 1000 experiments have a confidence interval which contains the mean of the population, we can say that there is 95% confidence interval.

LLN: It suggests that as sample size increases, the sample mean starts to converge to the population mean.

Let us consider $\sigma = 2$, then

n = sample size	$\sigma/\sqrt{n}$ = Standard error
20	0.4472
50	0.2828
100	0.2
200	0.1414
1000	0.0632

We can see that the spread of the histogram of sample means becomes narrower as n increases. In above case from histogram I can see at n = 20, the sample mean is between -1.5 to 1.5

to when n = 1000, the sample mean is between -0.2 to 0.2

CLT: Since the population has a gaussian distribution, the sample mean will also have gaussian distributions.

Sample mean: The sample means should converge as n increases.

we know that $Var(\bar{X}_n) = \frac{\sigma^2}{n} = \frac{4}{n}$ $(\sigma = 2)$

At $n = 20$, $Var(\bar{X}_{20}) = 0.2$ & at $n = 1000$, $Var(\bar{X}_{1000}) = 0.004$.

In the plots we can see that the value of the empirical mean approaches the population mean, from the histograms, we have

n	Empirical mean
20	0.0101
50	0.0046
100	0.0066
200	0.0023
1000	0.0028

In the plot of confidence intervals, the lower and upper confidence limits scatter around the mean starts to concentrate towards the true mean as the value of n increases.

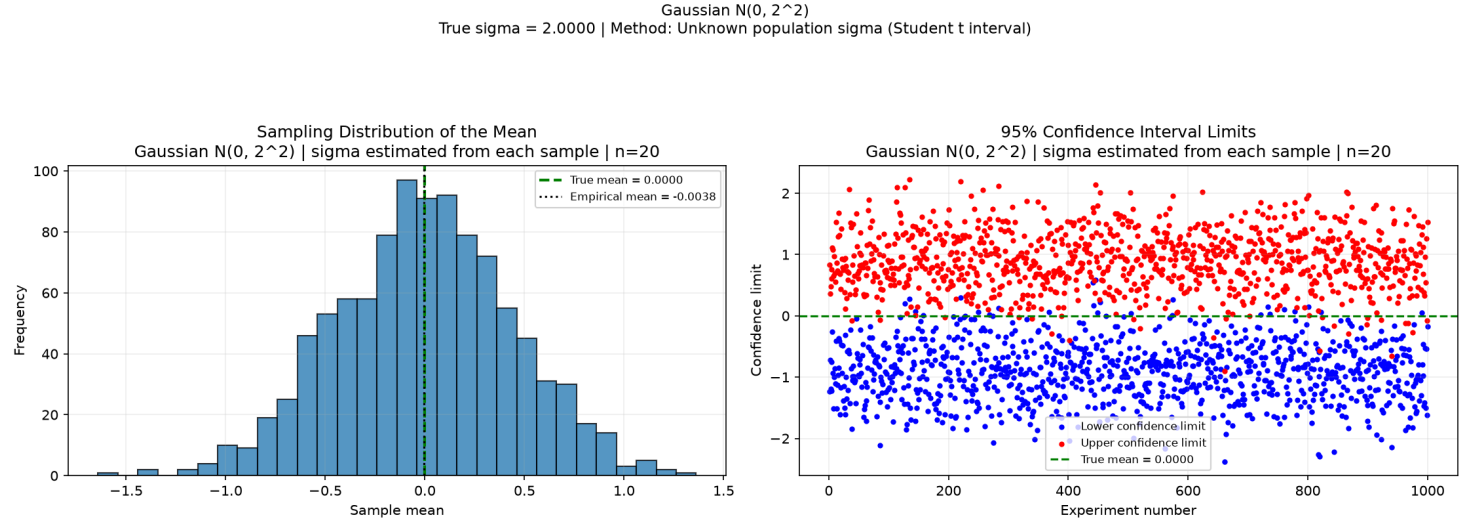
No. of times true mean was within the confidence interval

σ \ n	20	50	100	200	1000
2	954	950	947	959	957
3	953	948	952	944	953
5	955	947	939	952	940

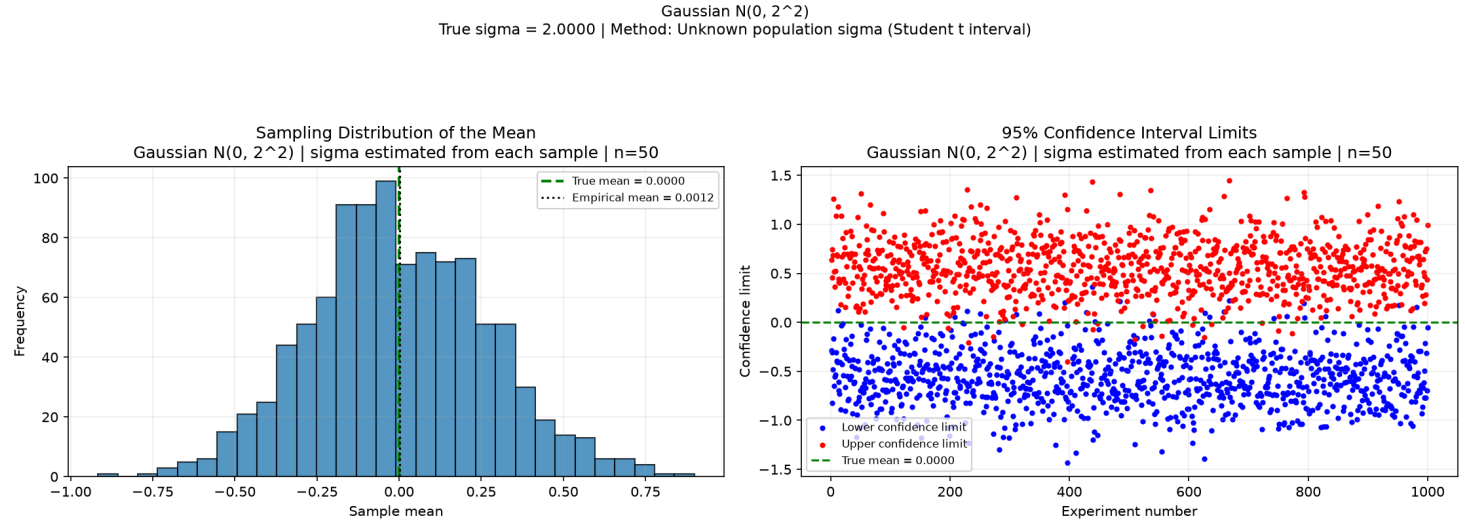
values are out of 1000

(b) Now pretend σ is not known and use sample variance as discussed in class in all of the above.

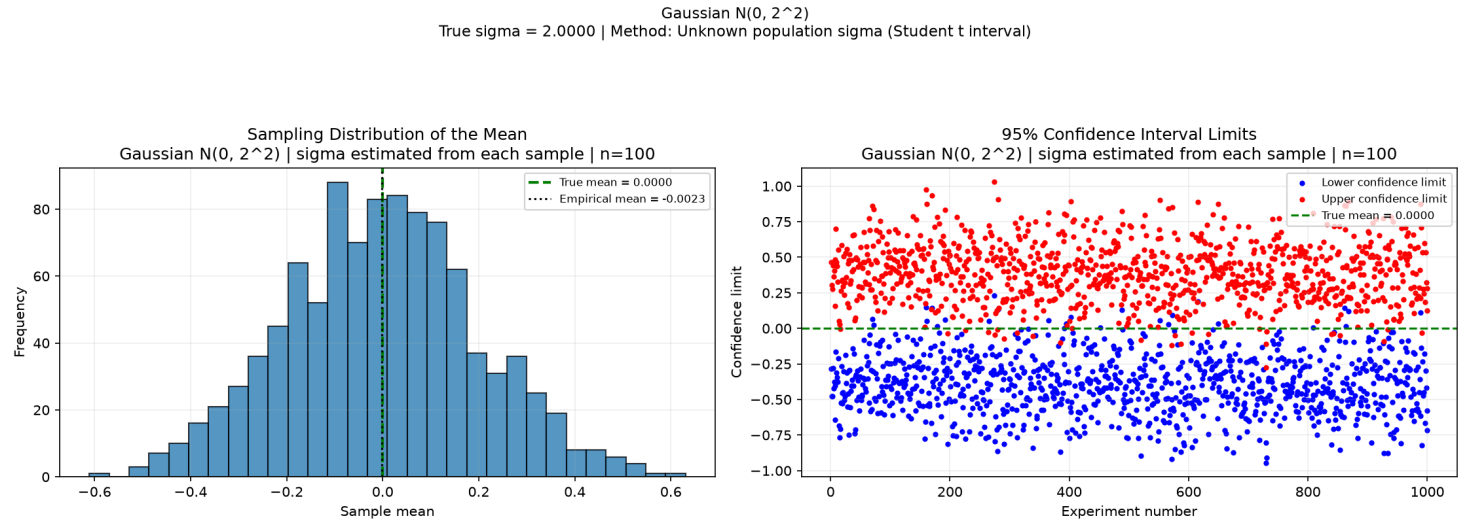
σ = 2 and n = 20



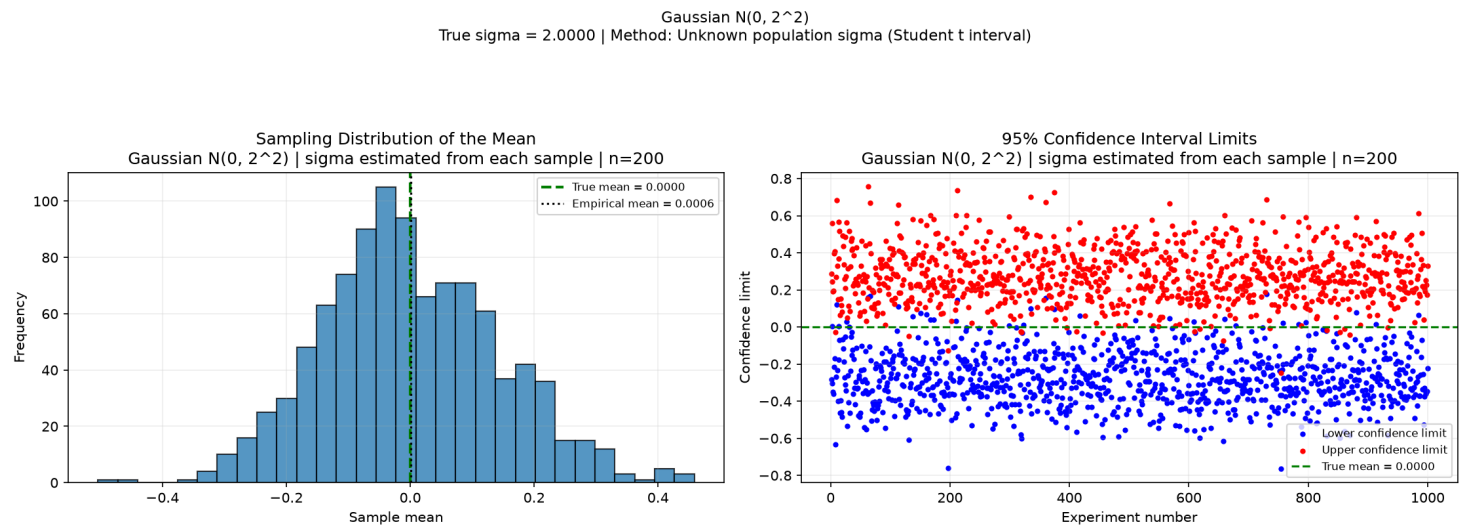
σ = 2 and n = 50



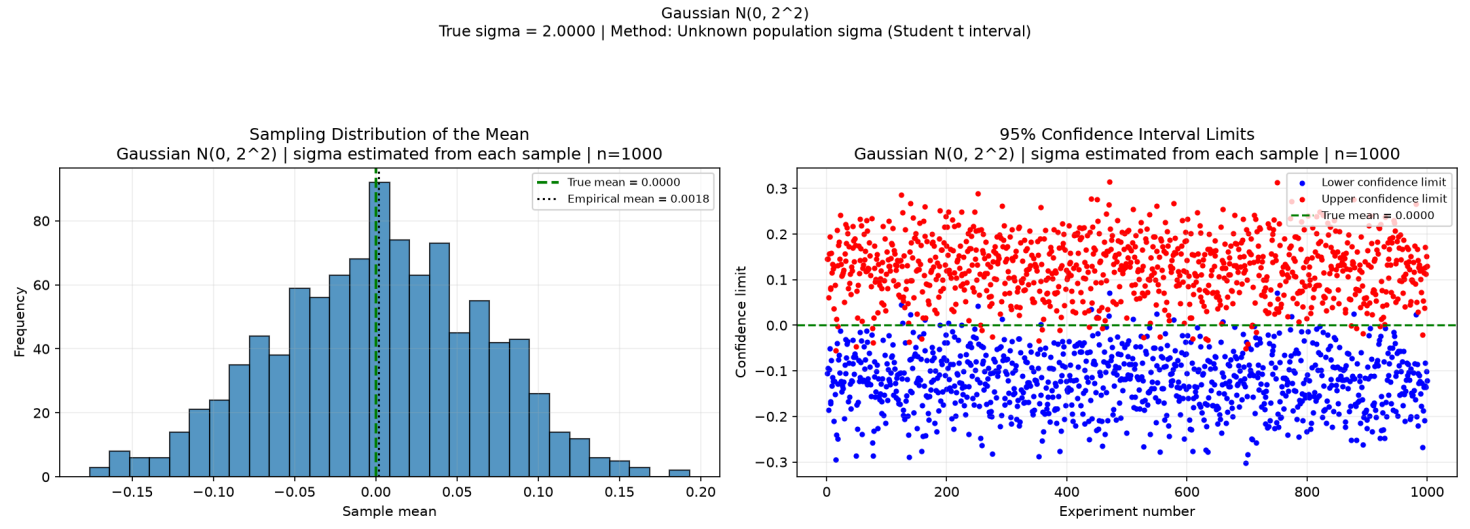
σ = 2 and n = 100



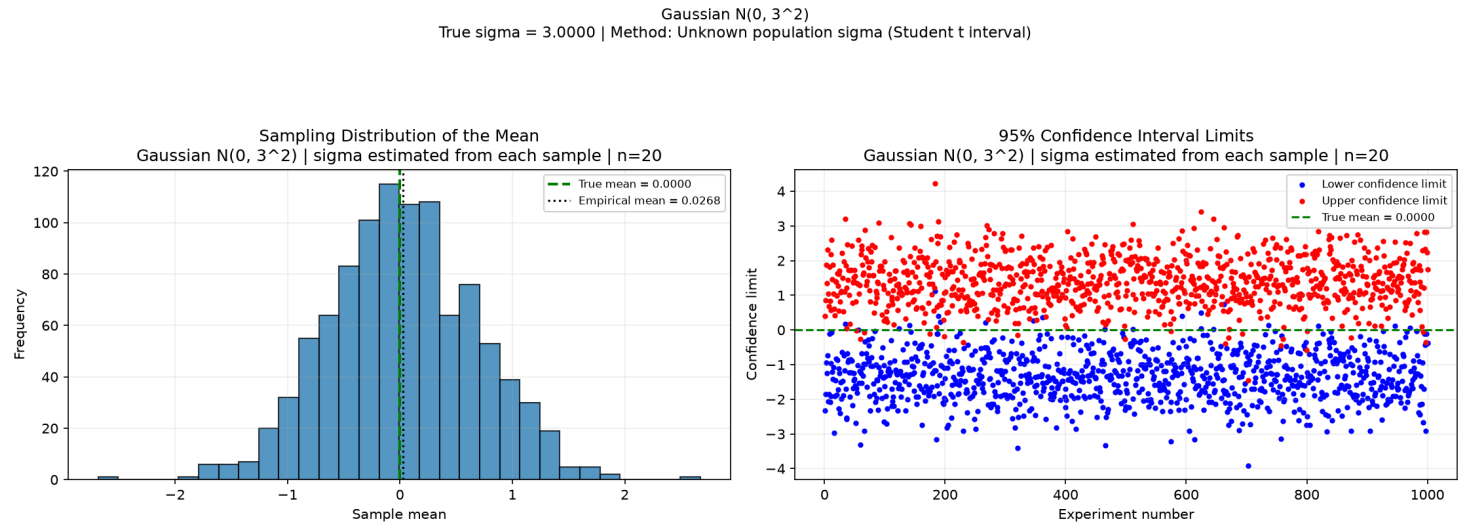
σ = 2 and n = 200



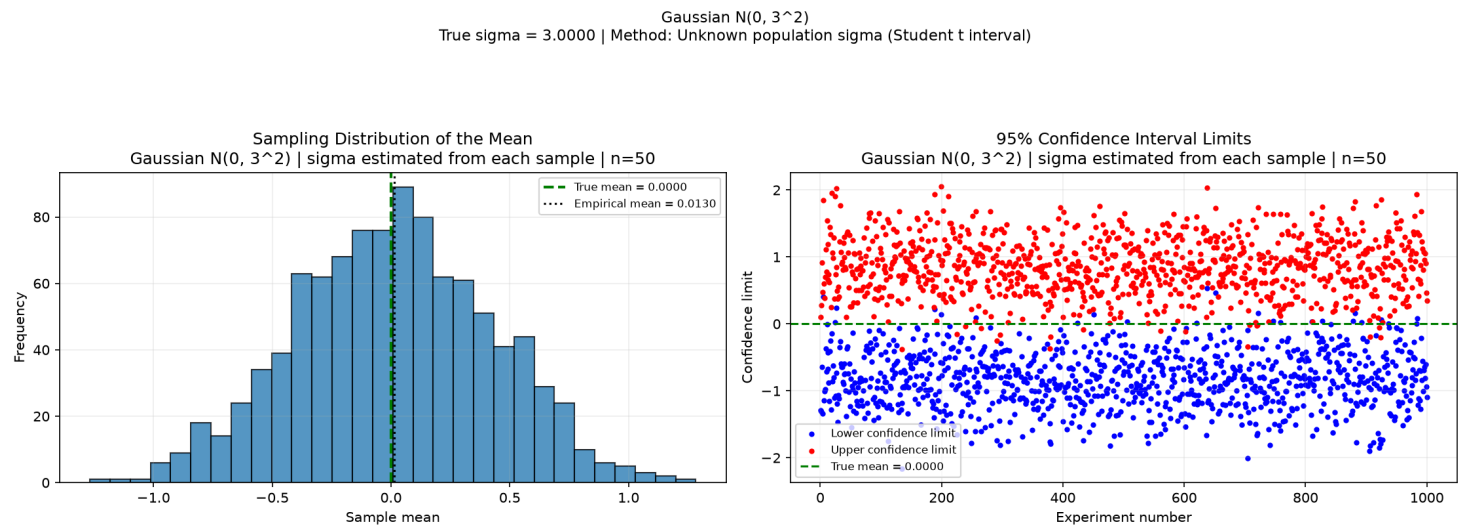
σ = 2 and n = 1000



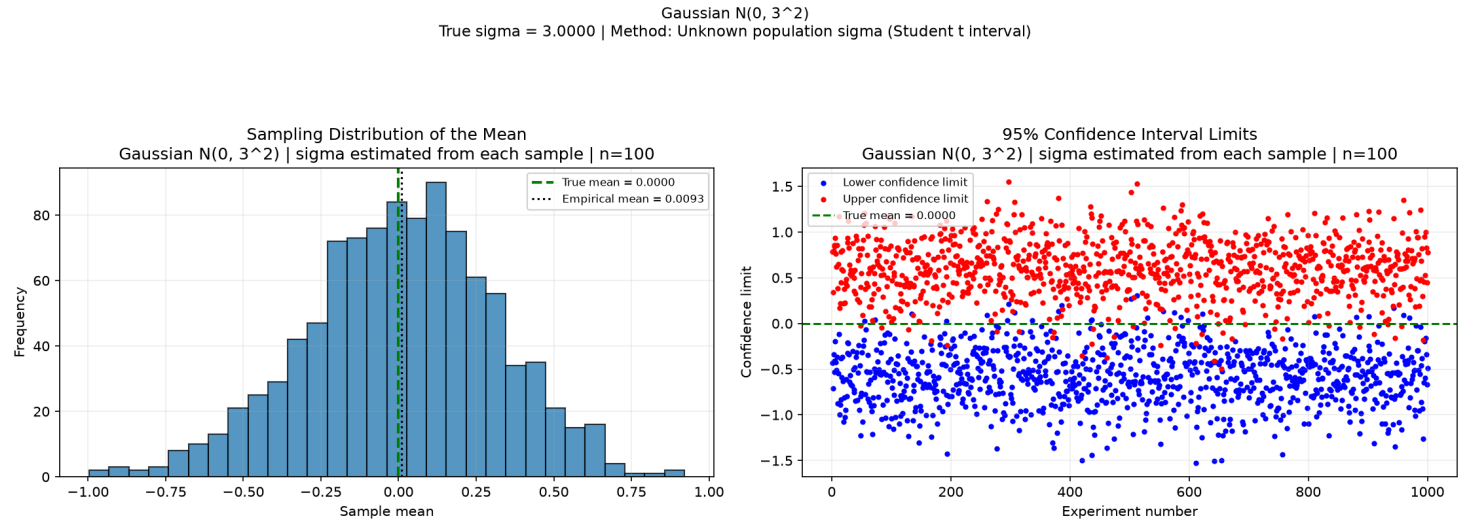
σ = 3 and n = 20



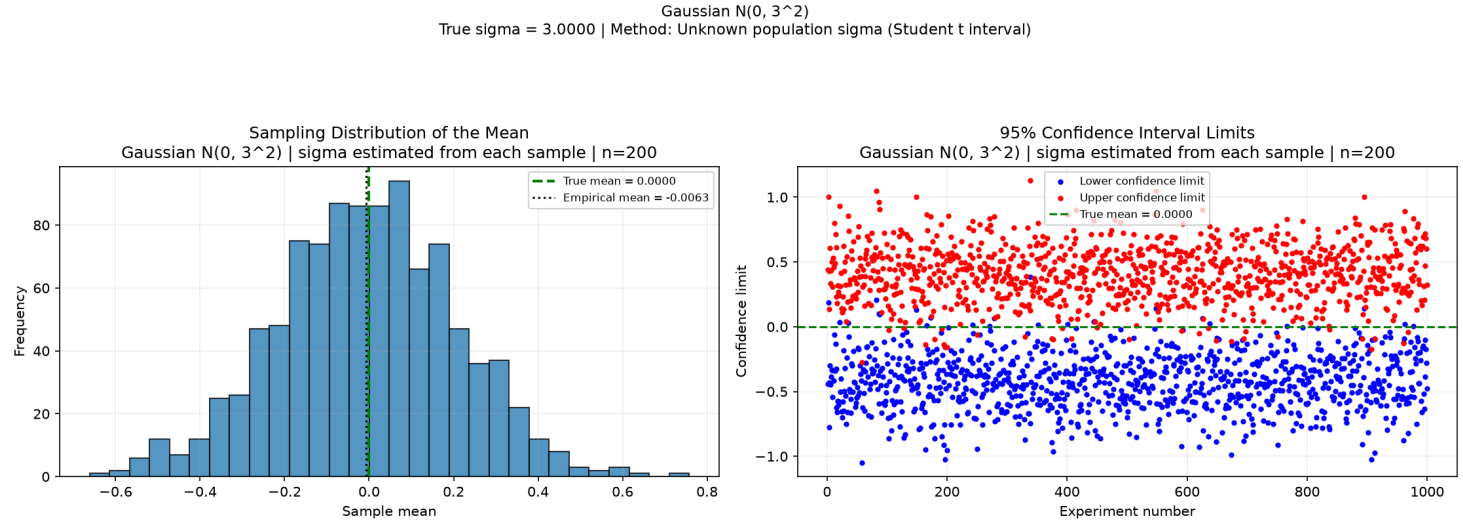
σ = 3 and n = 50



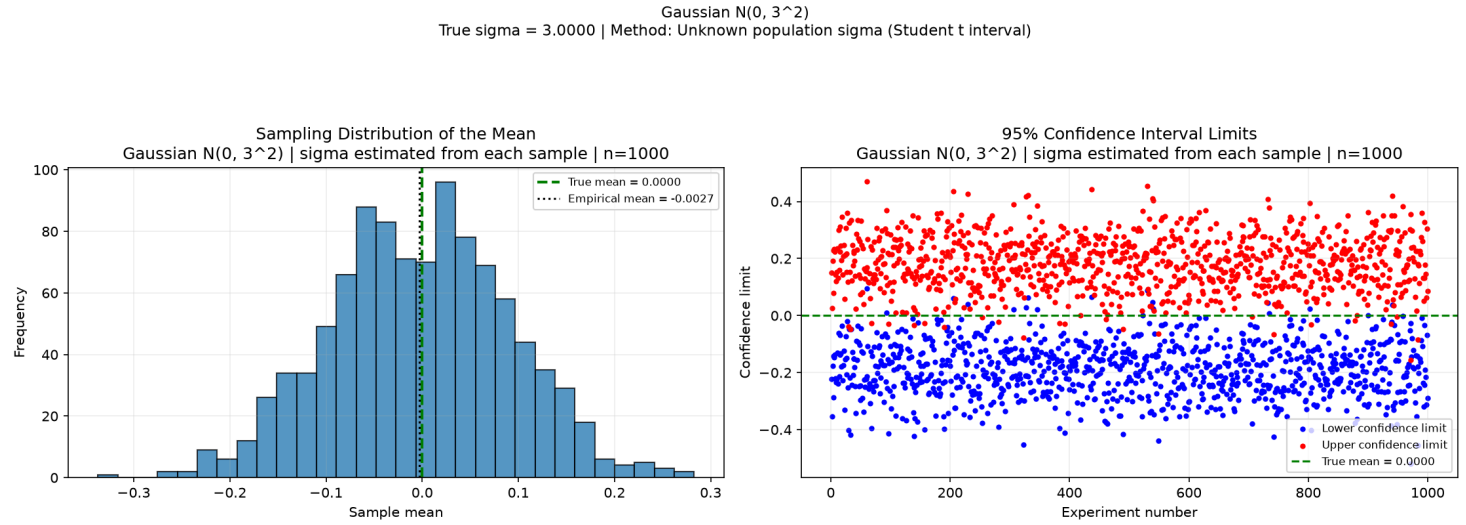
σ = 3 and n = 100



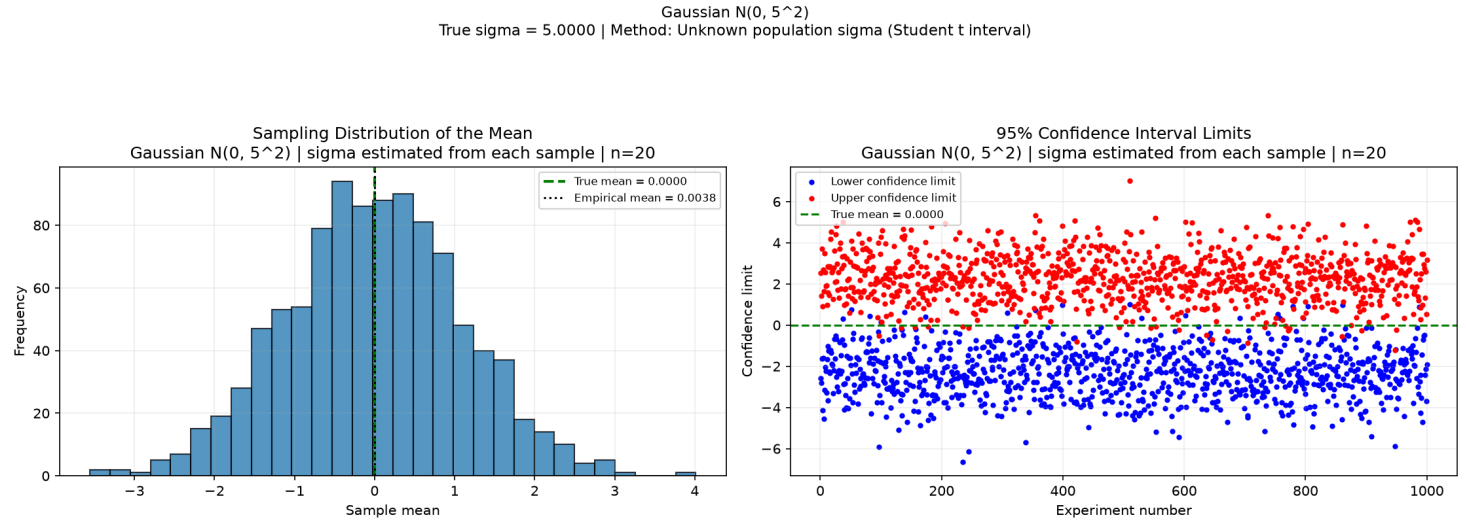
σ = 3 and n = 200



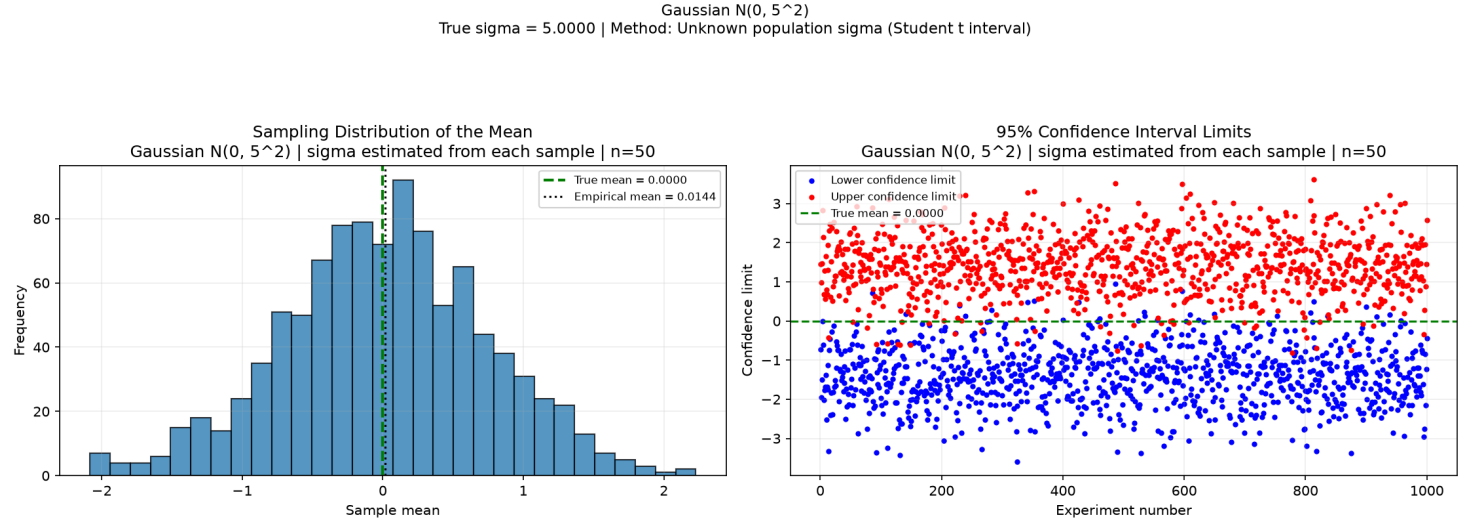
σ = 3 and n = 1000



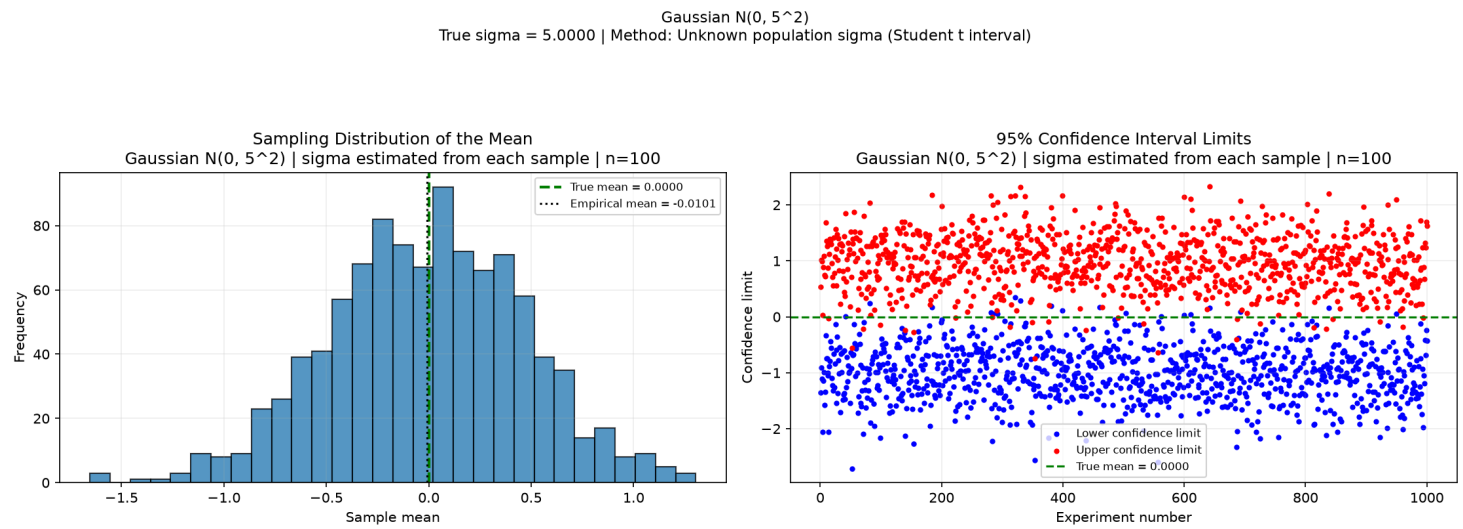
σ = 5 and n = 20



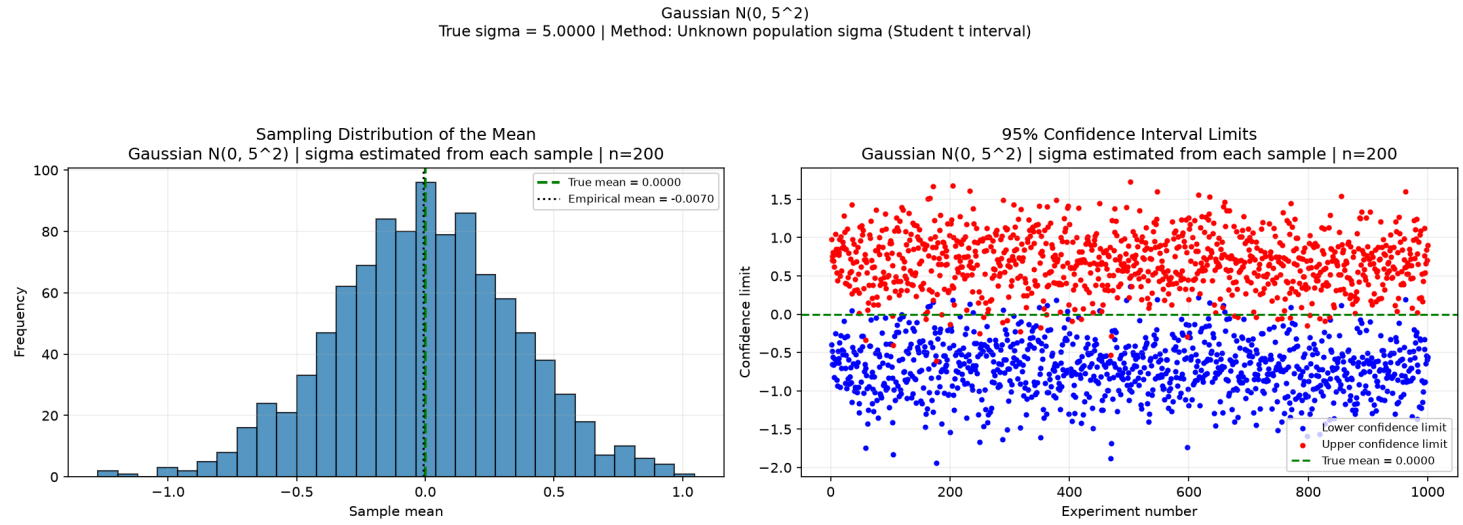
σ = 5 and n = 50



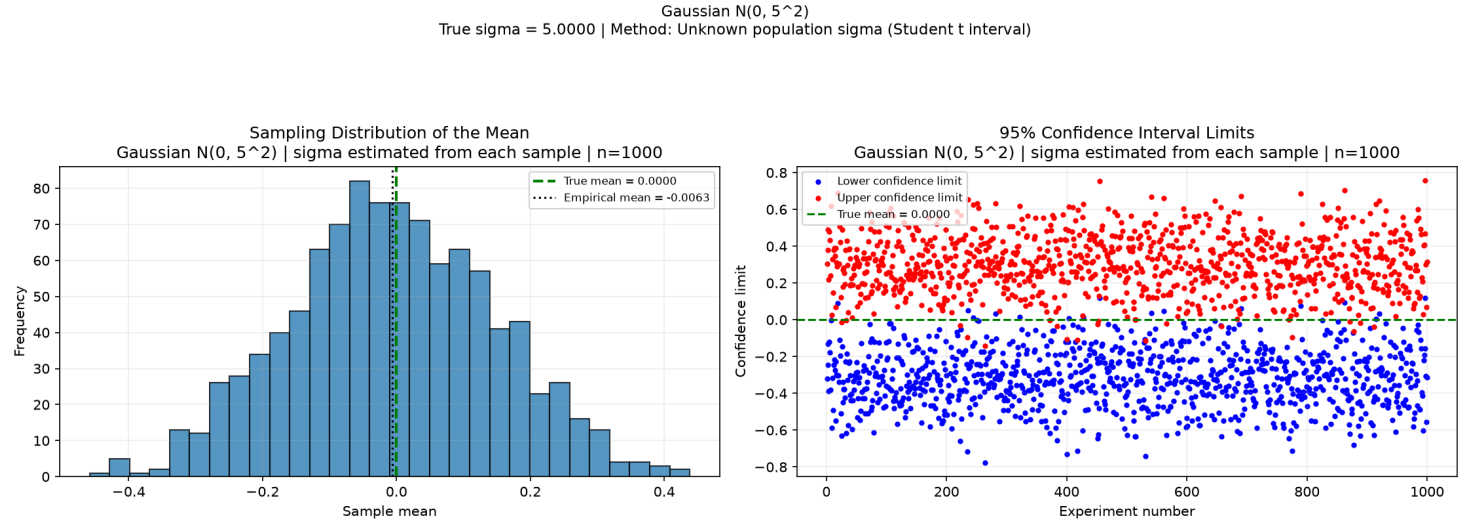
σ = 5 and n = 100



σ = 5 and n = 200



σ = 5 and n = 1000



(b) Gaussian with unknown sigma

The underlying experiment is similar to part(a) with one main distinction, i.e. the σ will be treated as unknown and it will be estimated using the sample standard deviation. So the iterations will be on $\sigma = 2, 3, 5$ & $n = 50, 20, 100, 200, 1000$ but for calculations we will be using the σ of the sample.

Let's look at some calculations :-

Consider $n = 20, \mu = 0, \sigma = 2$ (Population σ, which is not known)

Sample mean, $\bar{X} = 0.4$ & Sample $\sigma = 2.1 = S$

→ Standard error

$$SE(\bar{X}) = \frac{S}{\sqrt{n}} = \frac{2.1}{\sqrt{20}} \approx 0.4696$$

Confidence interval = Point estimate ± Margin of error

= Point estimate ± (Critical value × Standard error)

$$= 0.4 \pm t_{0.975, 19} \times 0.4696$$

Since σ is unknown, we use the t-distribution with $n - 1 = 19$ degrees of freedom

$$= 0.4 \pm 2.093 \times 0.4696$$

$$= 0.4 \pm 0.9829$$

$$[-0.5829, 1.3829]$$

Since 0 lies in the interval, it covers the true population mean. We can see that the coverage may not always be 95% and it may not always increase unlike the other parameters.

LLN: The observation is similar to part (a) where the histograms of the sample means get narrower around 0 as n increases.

Note:- Although the confidence interval uses an estimated standard deviation(S), The theoretical spread of the sample means is determined by the population σ.

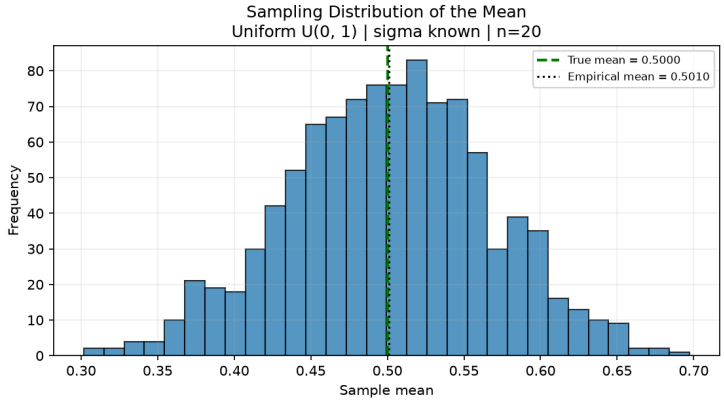
CLT: A gaussian distribution will have a gaussian distribution for its sample means. The histograms have the bell shape similar to (a).

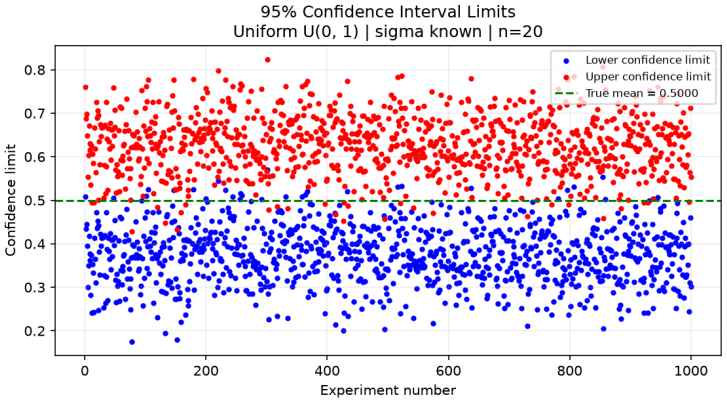
Sample mean: Similar trends in the graphs can be seen as (a). The sample mean starts to converge as n increases. Even though the confidence intervals use the t-distribution rather than the standard normal distribution, the intervals get narrower as n increases. Their widths may vary between repetitions because of no fixed deviation for each experiment.

(c) Now repeat part 1 above for three cases of F being uniformly distributed: (0,1), (−1,1), and (−2,2).

U(0,1) at n=20

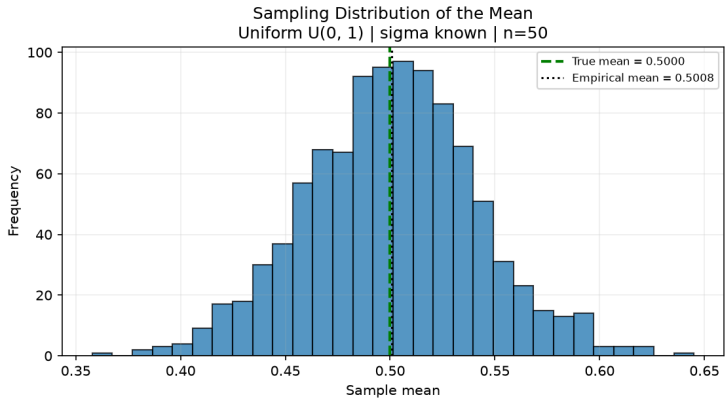
Uniform U(0, 1)
True sigma = 0.2887 | Method: Known population sigma (normal z interval)

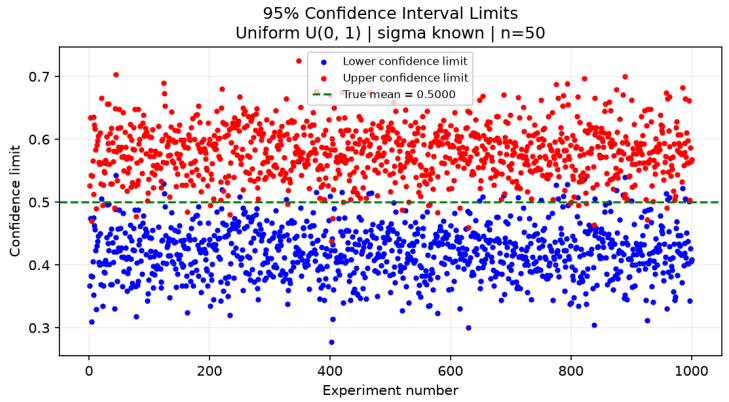




U(0,1) at n=50

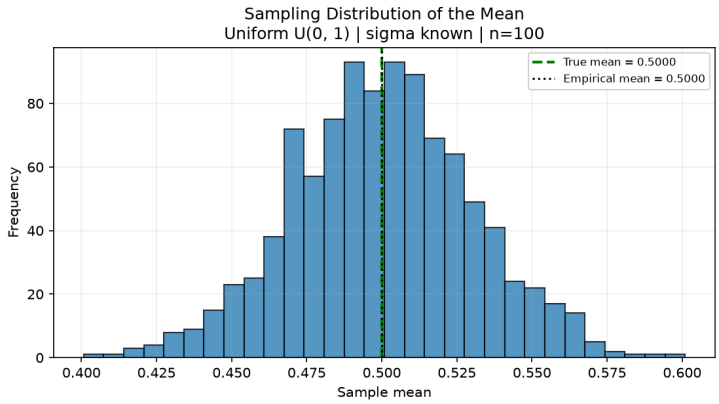
Uniform U(0, 1)
True sigma = 0.2887 | Method: Known population sigma (normal z interval)

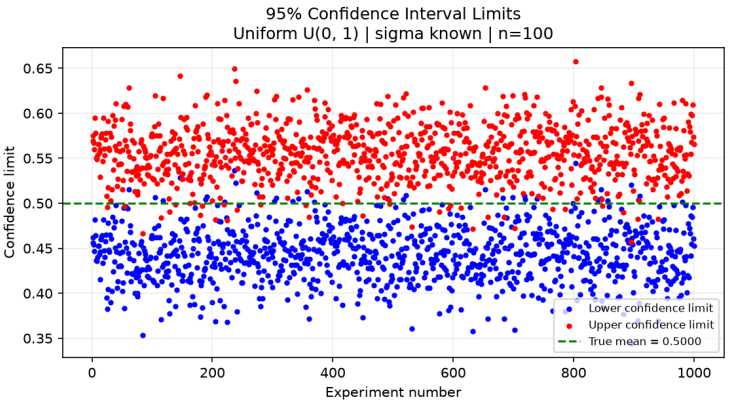




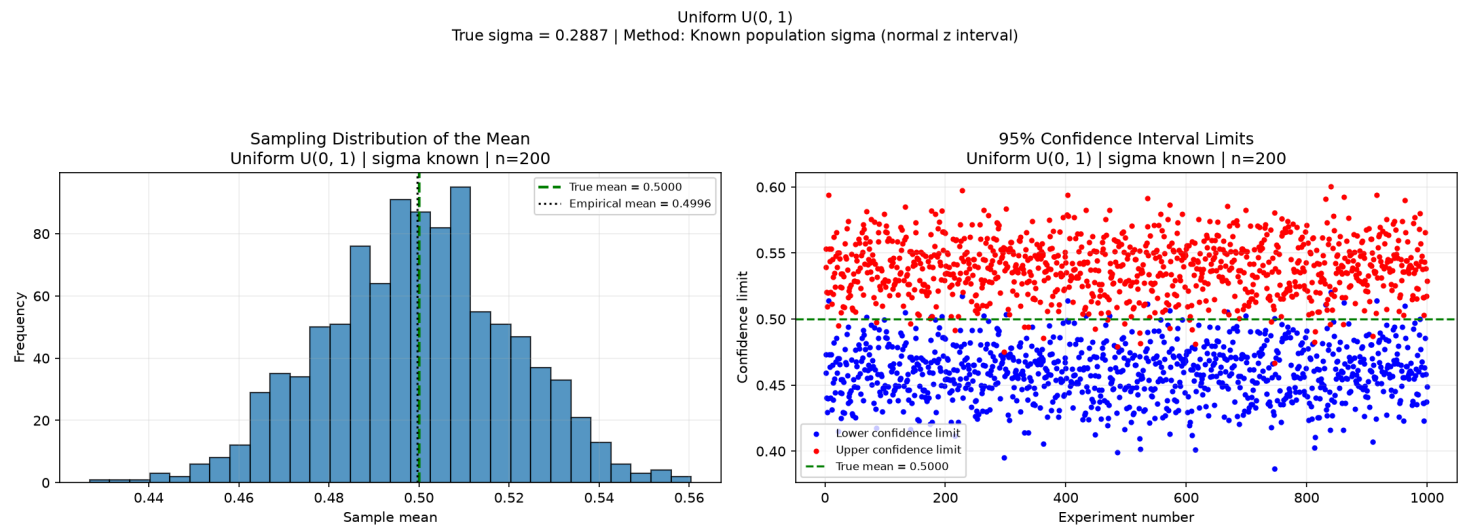
U(0,1) at n=100

Uniform U(0, 1)
True sigma = 0.2887 | Method: Known population sigma (normal z interval)

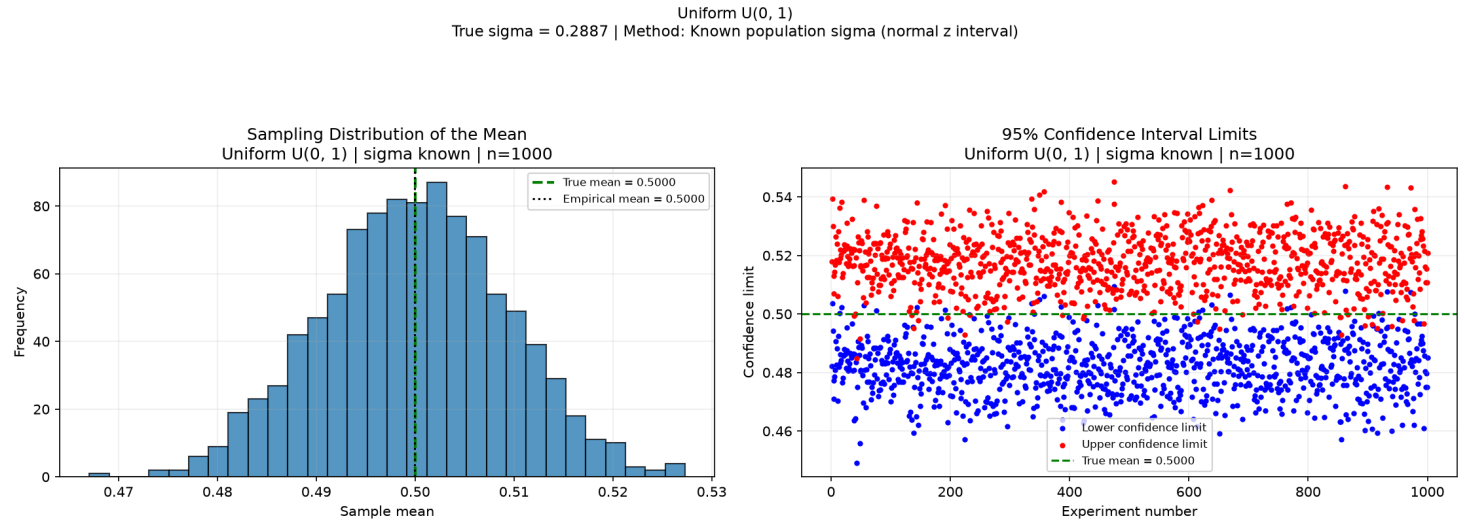




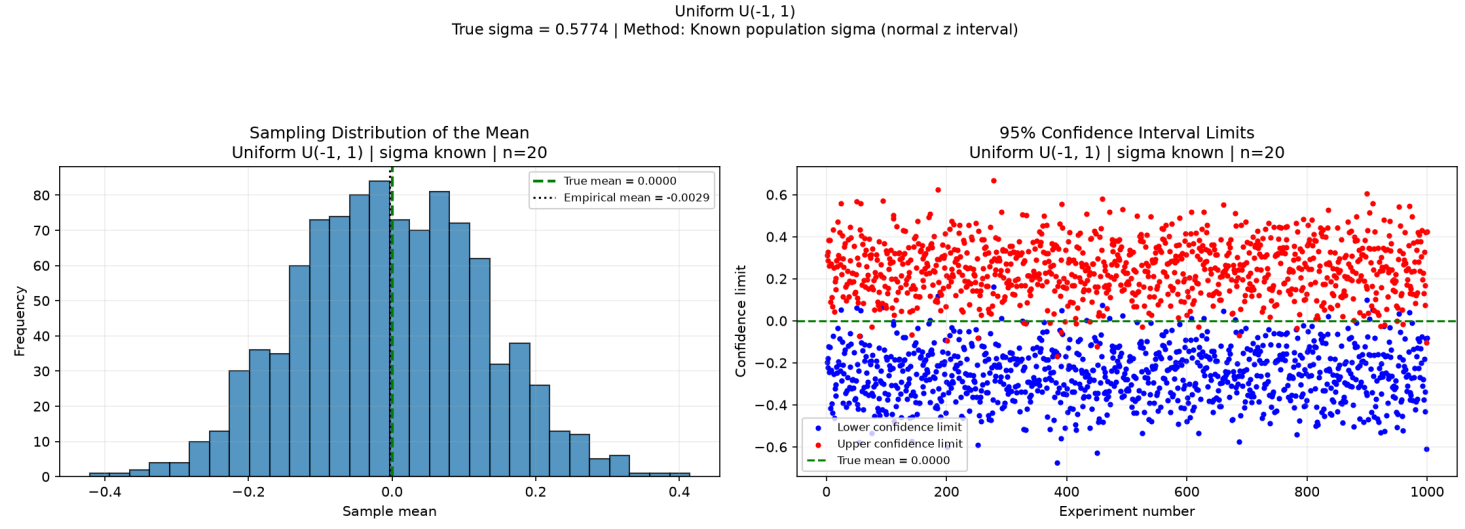
U(0,1) at n=200



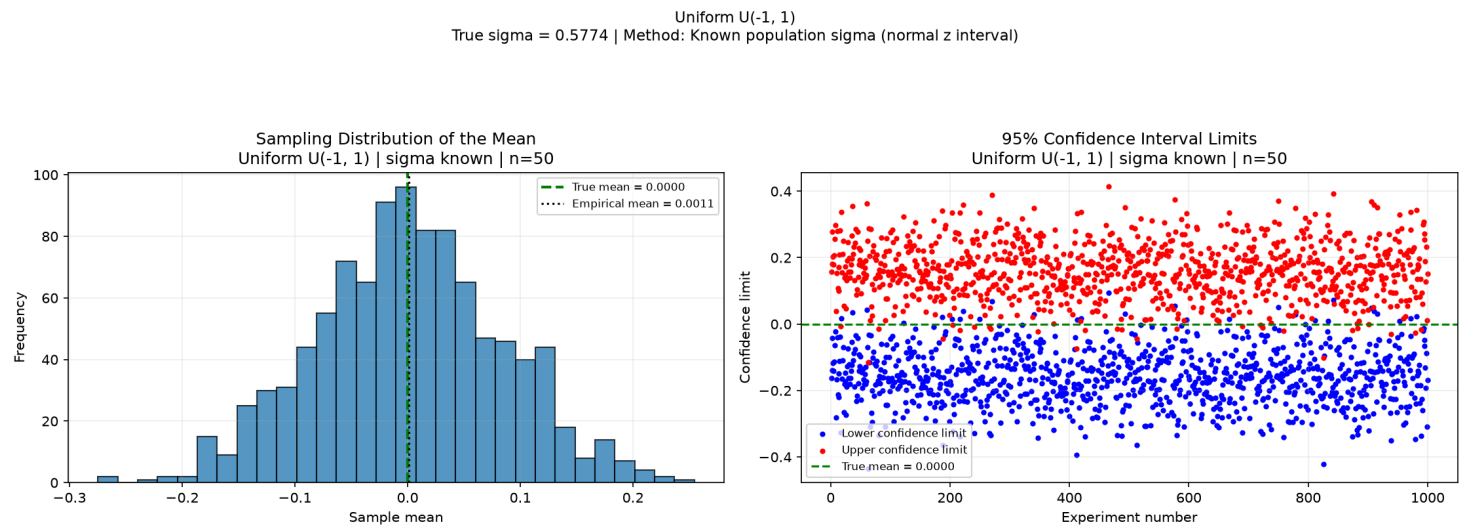
U(0,1) at n=1000



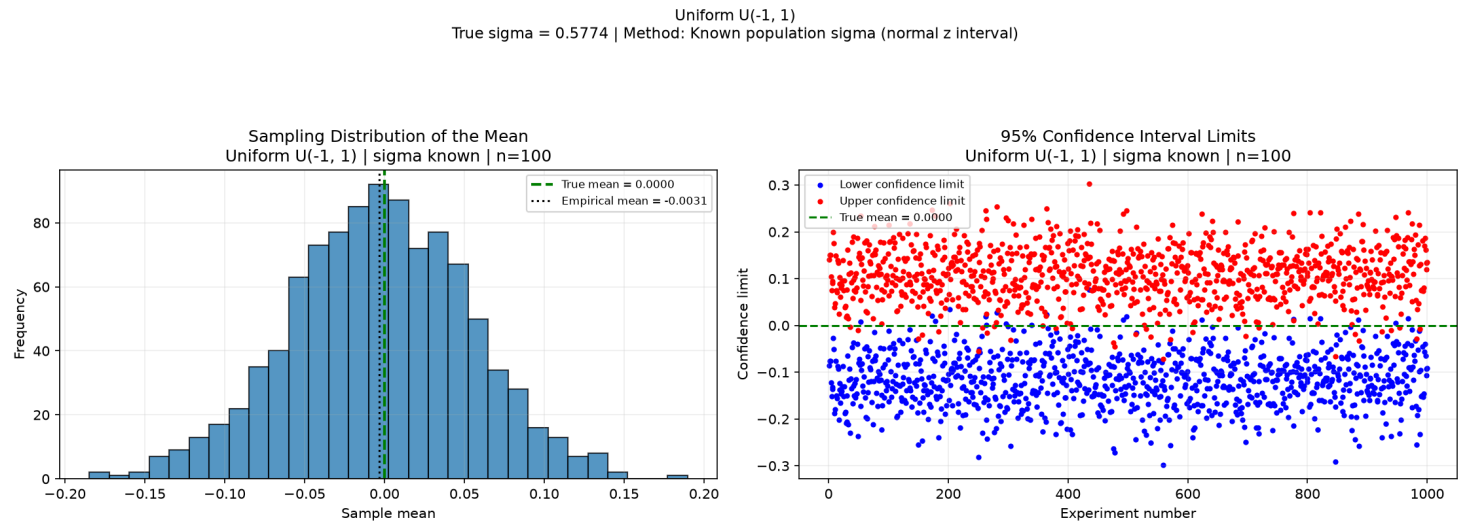
U(-1,1) at n=20



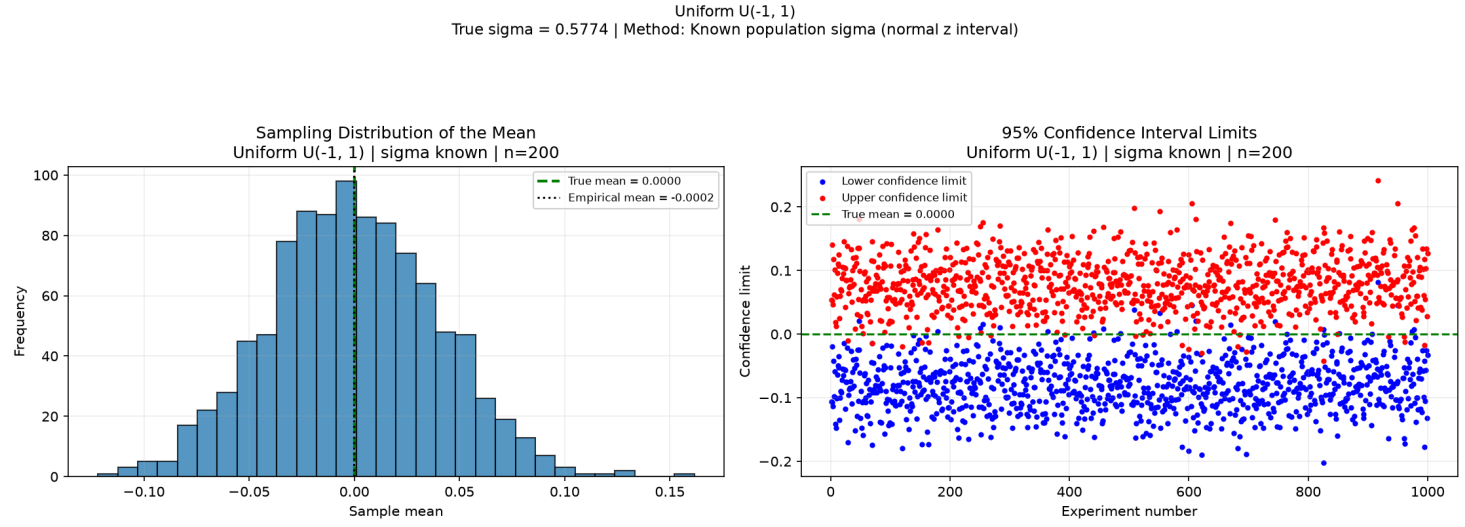
U(-1,1) at n=50



U(-1,1) at n=100

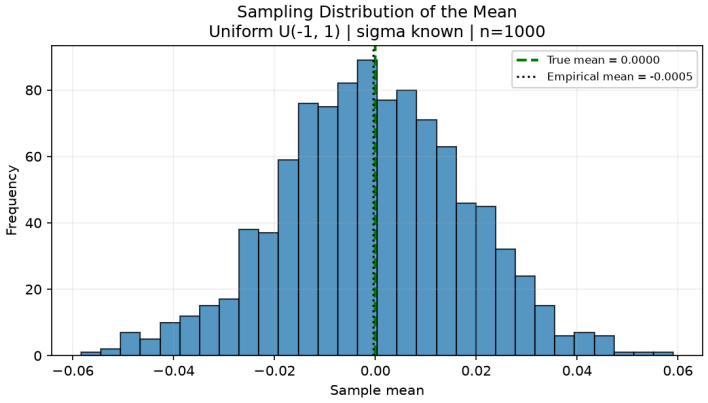


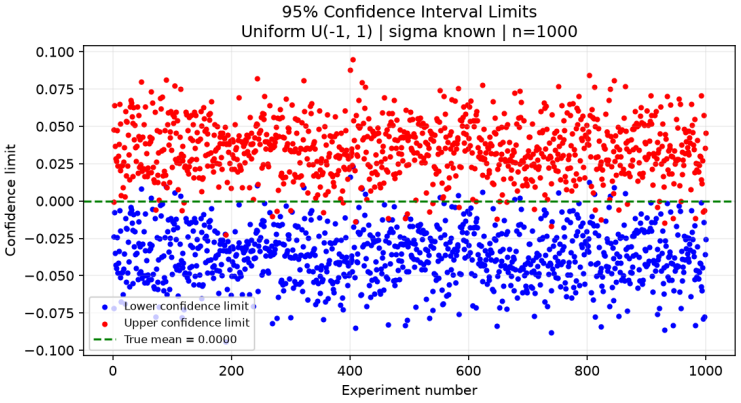
U(-1,1) at n=200



U(-1,1) at n=1000

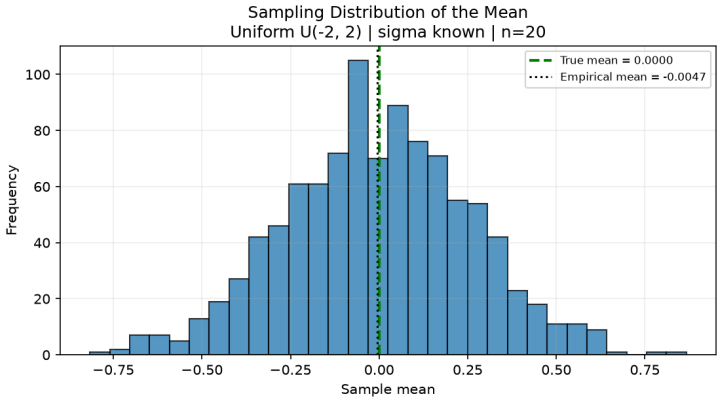
Uniform U(-1, 1)
True sigma = 0.5774 | Method: Known population sigma (normal z interval)

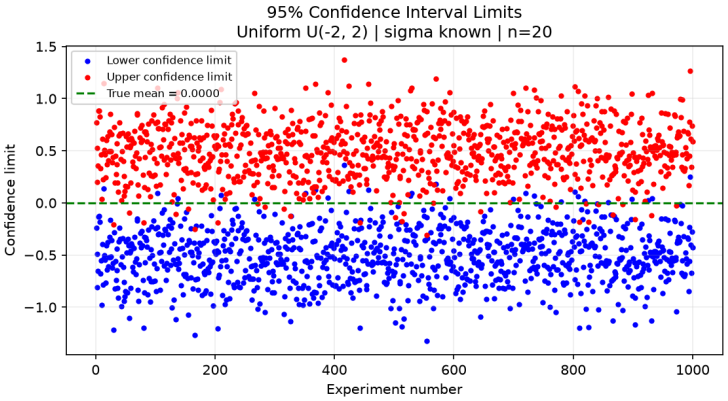




U(-2,2) at n=20

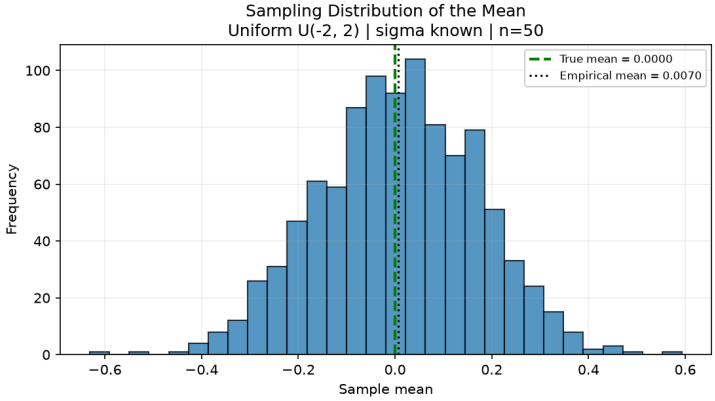
Uniform U(-2, 2)
True sigma = 1.1547 | Method: Known population sigma (normal z interval)

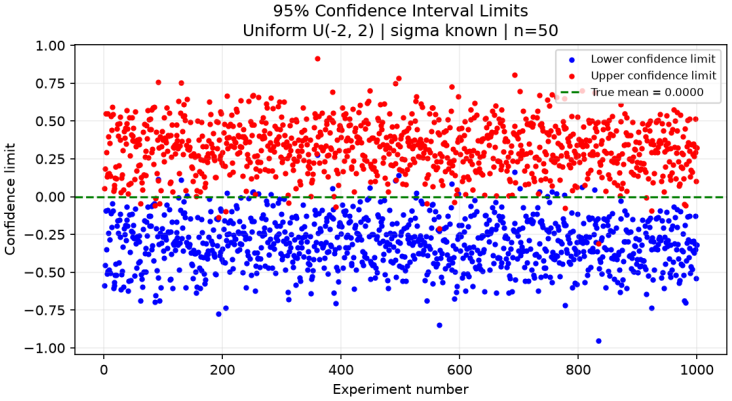




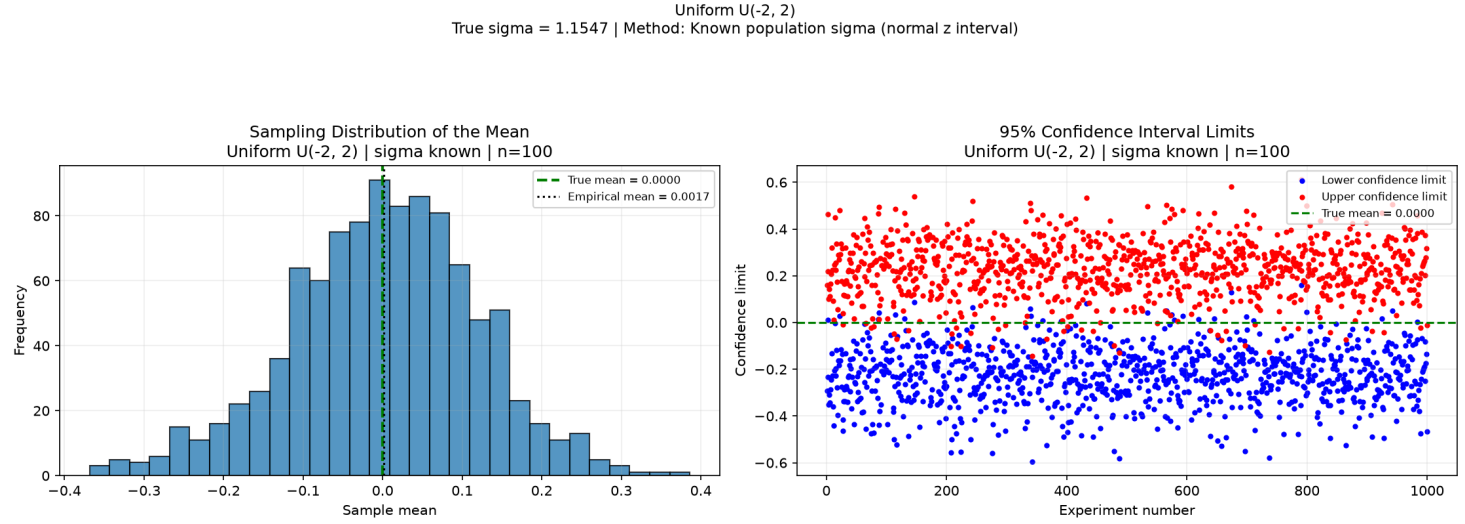
U(-2,2) at n=50

Uniform U(-2, 2)
True sigma = 1.1547 | Method: Known population sigma (normal z interval)

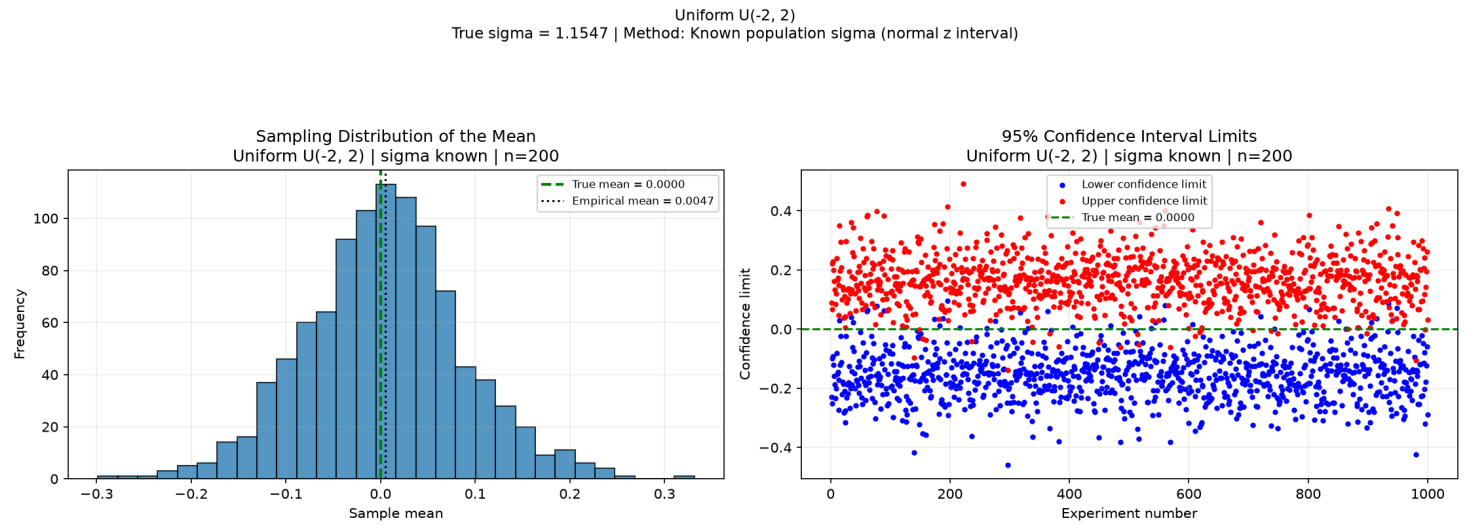




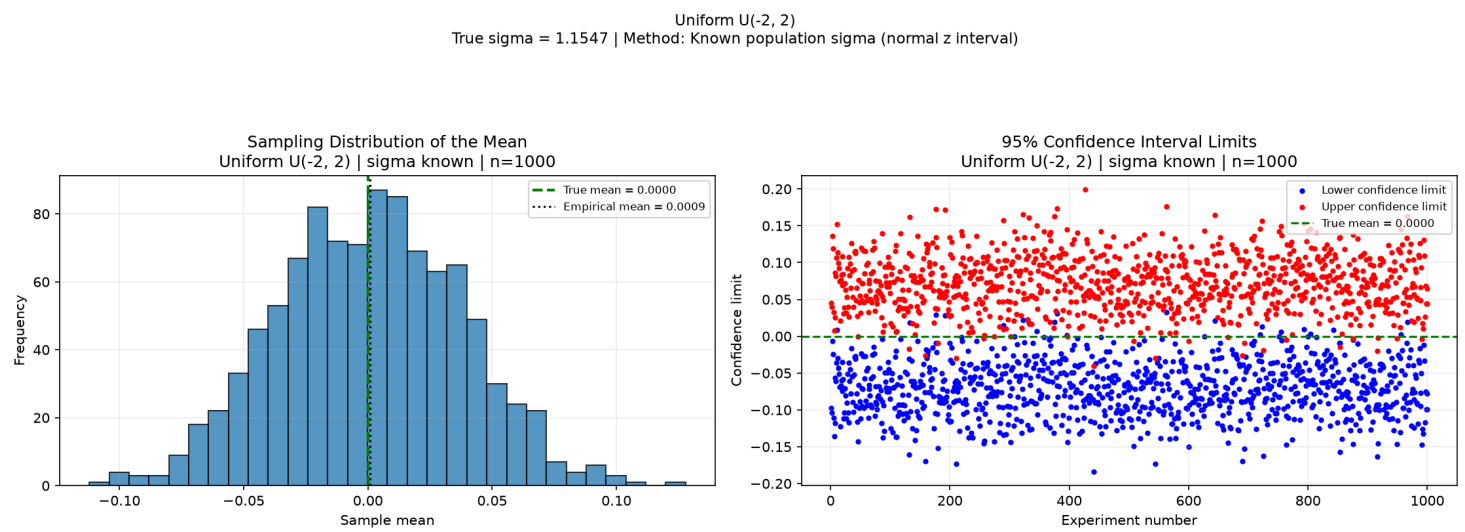
U(-2,2) at n=100



U(-2,2) at n=200



U(-2,2) at n=1000



(c) Uniform distribution

The experiment is similar to (a), the difference is in how the mean & std deviation are calculated for a uniform distribution. In this case there are 3 uniform distributions

$X \sim U(0,1)$

$X \sim U(-1,1)$

$X \sim U(-2,2)$

We do the same exercise of taking samples of $n = 20, 50, 100, 200$ & 1000. Each of this exercises is repeated 1000 times based on the fact that are iid's.

Let's see calculations for one of them :-

for $U(a,b)$ $\mu = \frac{a+b}{2}$, $\sigma = \sqrt{\frac{(b-a)^2}{12}}$

then we have $U(0,1)$ &, $\mu = \frac{0+1}{2} = 0.5$

$$\sigma = \sqrt{\frac{(1-0)^2}{12}} \approx 0.2887$$

These would be the values for the population.

Let's assume for one of the experiments, sample mean is 0.46

$$\bar{X} = 0.46$$

→ Standard error

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{0.2887}{\sqrt{20}} \approx 0.0645$$

Considering the confidence interval will be a normal distribution, we have

$$\text{Confidence interval} = \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}$$

$$= 0.46 \pm 1.96 \times 0.0645$$

$$= 0.46 \pm 0.1264$$

$$[0.3336, 0.5864]$$

The true mean 0.5 is in this confidence interval.

From the plots we can see that (coverage results) unlike the gaussian distribution the confidence interval is slightly lesser especially for smaller sample sizes.

LLN: As sample size increases, sample mean should converge to population mean. (for $U(0,1)$)

Sample size n	Standard error $\sigma/\sqrt{n}$	Empirical mean
20	0.0645	0.5010
50	0.0408	0.5008
100	0.0289	0.5000
200	0.0204	0.4996
1000	0.0091	0.5000

The standard error reduces as n increases, in the histograms we can see the value of sample mean concentrated around 0.5 for $U(0,1)$ & around 0 for $U(-1,1)$ & $U(-2,2)$.

The values are very close to the population mean as n increases, even exact in 2 cases.

CLT: It states that standardized sample mean is approximately normally distributed:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

Also $\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$

for U(0,1) ~~$\bar{X}_n \approx N\left(0.5, \frac{1}{\sqrt{12n}}\right)$~~ $\bar{X}_n \approx N\left(0.5, \frac{1}{12n}\right)$

as the value of n increases, you can see that the variance value will decrease which means the values of the sample mean will be closely present with the mean as n increases. 0.5 in this case. 0 in the case of U(-1, 1) & U(-2, 2)

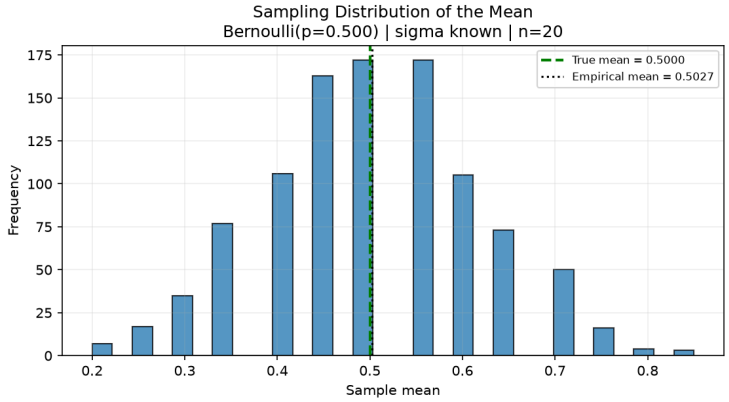
Sample mean:

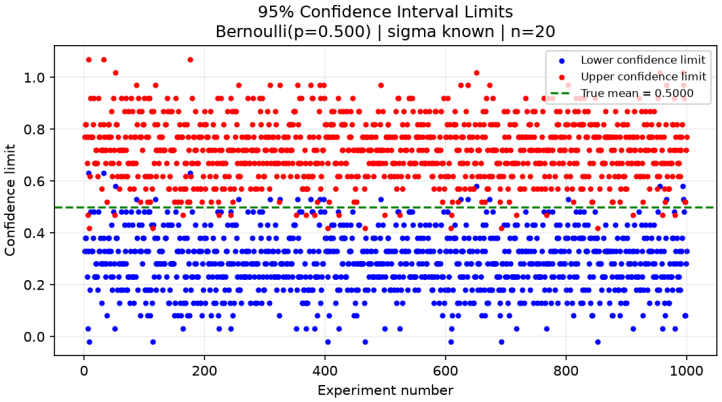
Similar to previous findings as n increases, even the values on the scatter plot become denser towards the mean of the distribution. (true mean)

(d) Finally repeat for F being a Bernoulli distribution

n=20

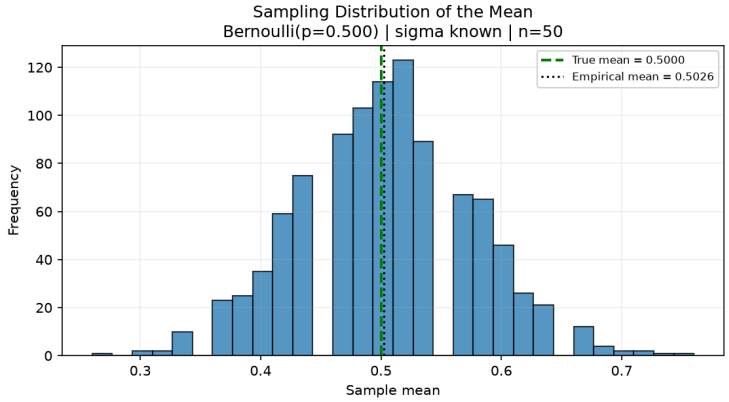
Bernoulli(p=0.500)
True sigma = 0.5000 | Method: Known population sigma (normal z interval)

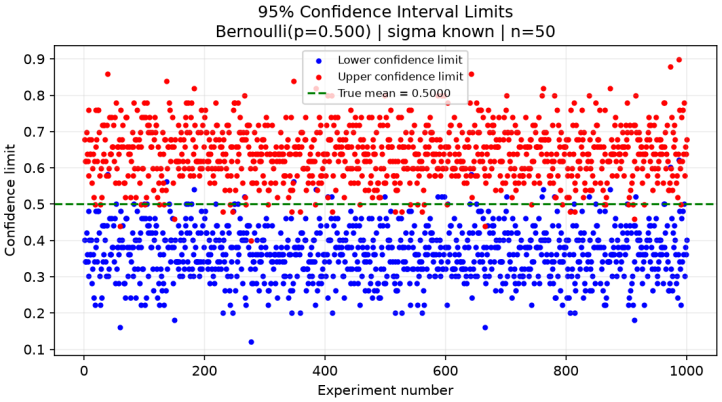




n=50

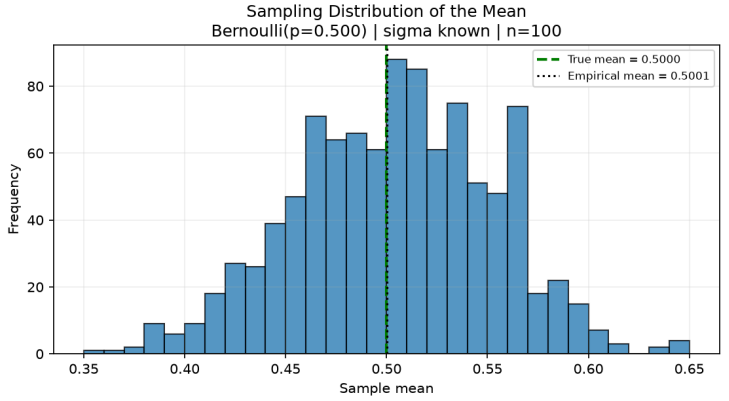
Bernoulli(p=0.500)
True sigma = 0.5000 | Method: Known population sigma (normal z interval)

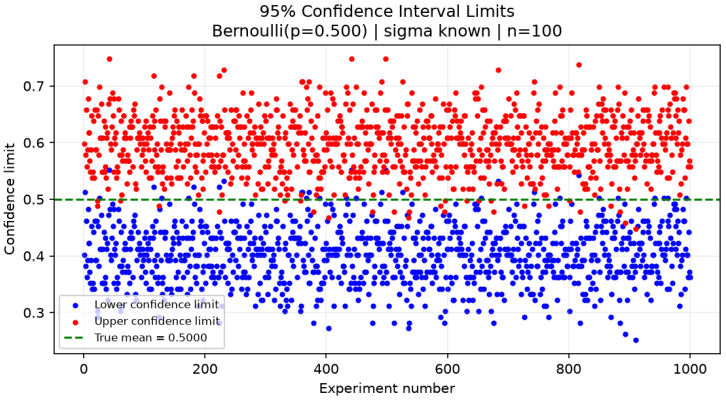




n=100

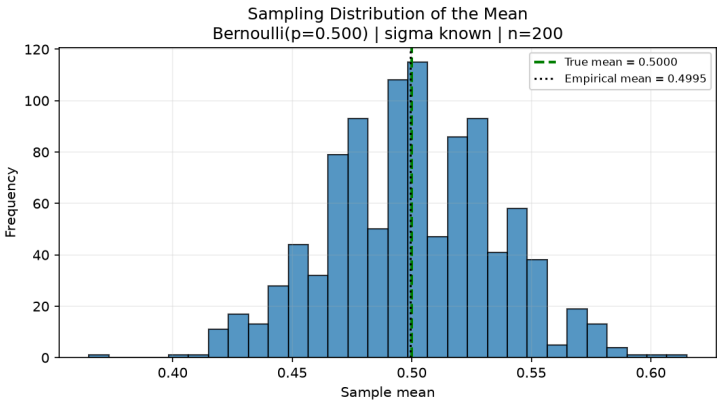
Bernoulli(p=0.500)
True sigma = 0.5000 | Method: Known population sigma (normal z interval)

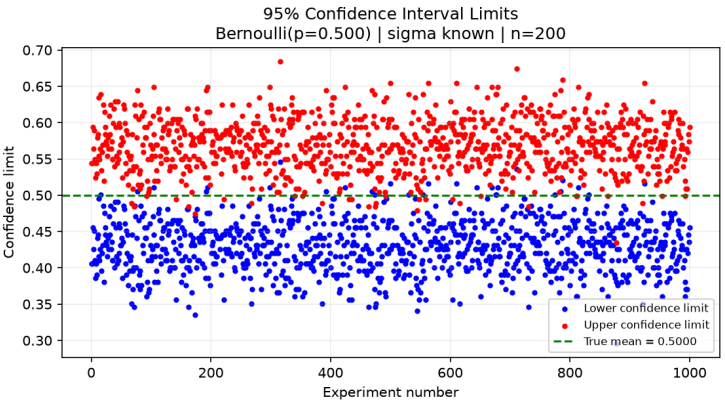




n=200

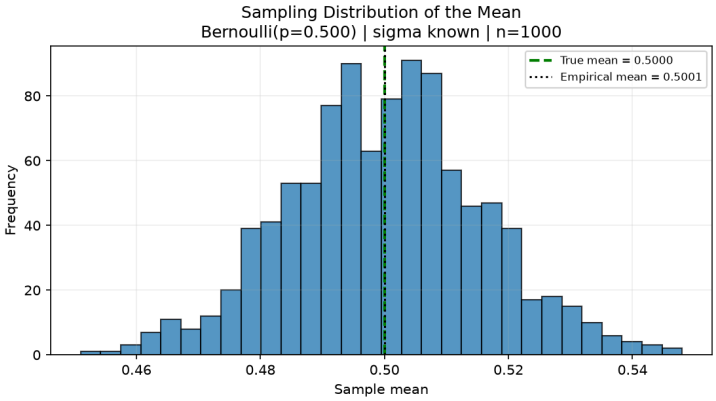
Bernoulli(p=0.500)
True sigma = 0.5000 | Method: Known population sigma (normal z interval)

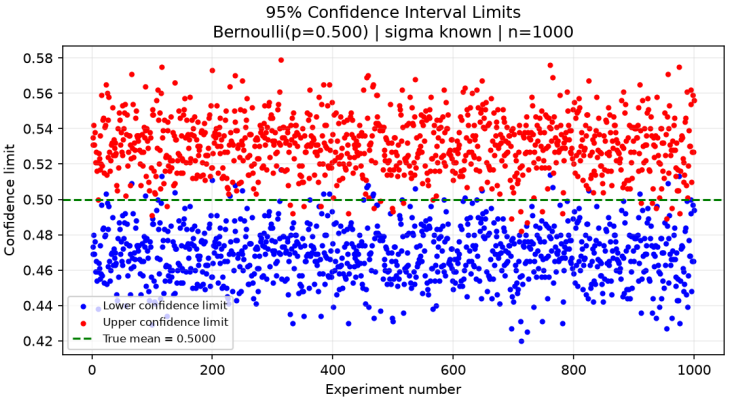




n=1000

Bernoulli(p=0.500)
True sigma = 0.5000 | Method: Known population sigma (normal z interval)





(d) Bernoulli distribution

Lets consider a Bernoulli distribution with $p = 0.5$,

$X = 1$ & $X = 0$ both have a probability of 0.5.

Now generate $n = 20, 50, 100, 200, 1000$ iid's & ~~repeat~~ calculate mean for each and repeat $k = 1000$ times.

For a single experiment

$\mu = E[X] = p$, $\sigma^2 = p(1-p)$

when $p = 0.5$

$\mu = 0.5$, $\sigma = 0.5$

Consider a case of $n = 20$ where there are 12 ones

Sample mean $= \bar{X} = \frac{12}{20} = 0.6$

→ Standard error

$$SE(\bar{X}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.5(0.5)}{20}} \approx 0.1118$$

Using normal approximation,

$$= \bar{X} \pm 1.96 \sqrt{\frac{p(1-p)}{n}}$$

$$= 0.6 \pm 1.96 \times 0.1118$$

$$= 0.6 \pm 0.2191$$

Confidence interval $= [0.3809, 0.8191]$

The true mean 0.5 lies within the interval in this case.

Similar to uniform distribution, the confidence interval might not always be

very close to 95%. especially for small sample sizes.

LLN: The trend of the sample mean getting ~~to~~ closer to population with increasing sample size continues even for Bernoulli distribution.

Sample size n	Standard error $\sigma/\sqrt{n}$
20	0.1118
50	0.0707
100	0.05
200	0.0354
1000	0.0158

CLT: With sufficiently large sample sizes, the standardized sample mean is approximately normally distributed.

$$\frac{\bar{X}_n - p}{\sqrt{p(1-p)/n}} \approx N(0,1)$$

$$\bar{X}_n \approx N\left(p, \frac{p(1-p)}{n}\right)$$

The bell shape or normalization of the sample means can be seen in the histogram at $n = 20$ it just gets better as n increases and the variance decreases.

Sample mean: Based on the variance, sample mean starts approaching true mean as sample size n starts increasing. This is evident from both the plots as their values start to converge & are densely populated towards the mean as n increases.

Common observations :-

The use of multiple distributions with different configurations has clearly brought out how increase in sample size n causes the sample mean to converge to population mean. ~~The~~ (LLN). The different distributions have sample mean histograms which are normally distributed despite it being based off of different types of distributions (CLT). The randomness of the selected iid's can be seen based on the fact that although the values of sample mean approaches population mean with increase in n, it is not always the case (there is also monotonicity or linear relationship), we just know that it gets closer.

2.Now consider the following.

(a) A manufacturing line produces 100 ohm resistors; The buyer wants a tolerance of 5% and willing to accept batch if the manufacturer can guarantee that at most 1% of the components are outside the

tolerance band.

(b) Assume Gaussian distribution for the resistance values;

(c) Compute the σ that the process should have.

2) It is a gaussian distribution, hence

$$X \sim N(\mu, \sigma^2)$$

$\mu = 100\,\Omega$ is the target mean resistance

The tolerance is 5% which would be $5\,\Omega$.

Therefore a resistor is acceptable if it is in below range

$$95 \leq X \leq 105$$

So the remaining range or values of the resistor should be 1% at most for the manufacturer to be selected by the buyer.

Then we have

$$P(X<95) + P(X>105) \leq 0.01$$

Since it is a gaussian distribution, it is symmetric around mean which means

$$P(X<95) = 0.005,\ P(X>105) = 0.005$$

The rest of the 99% resistors should be within the tolerance band.

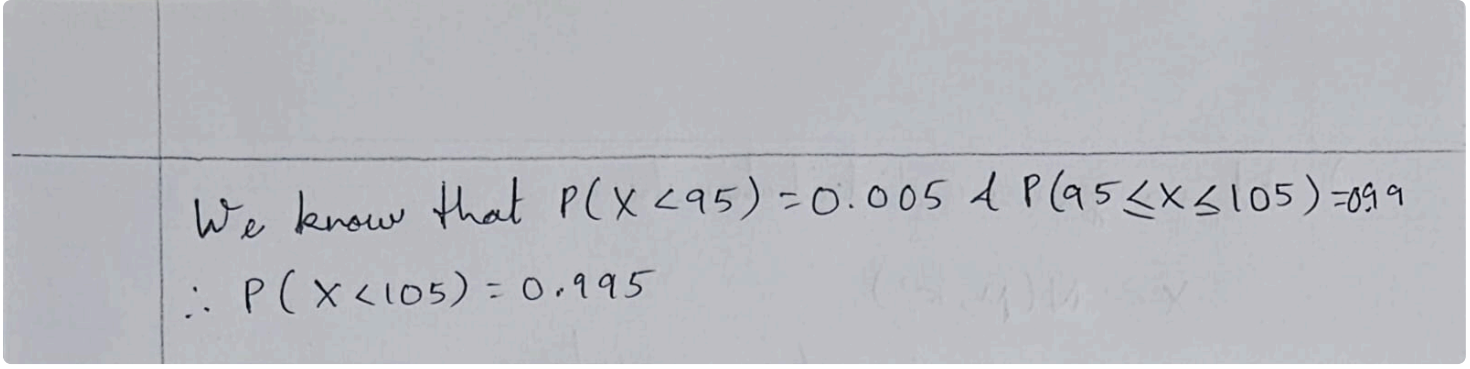
Calculate standard deviation

$$z = \frac{X-\mu}{\sigma}$$

$$z_{0.995} \approx 2.5758$$

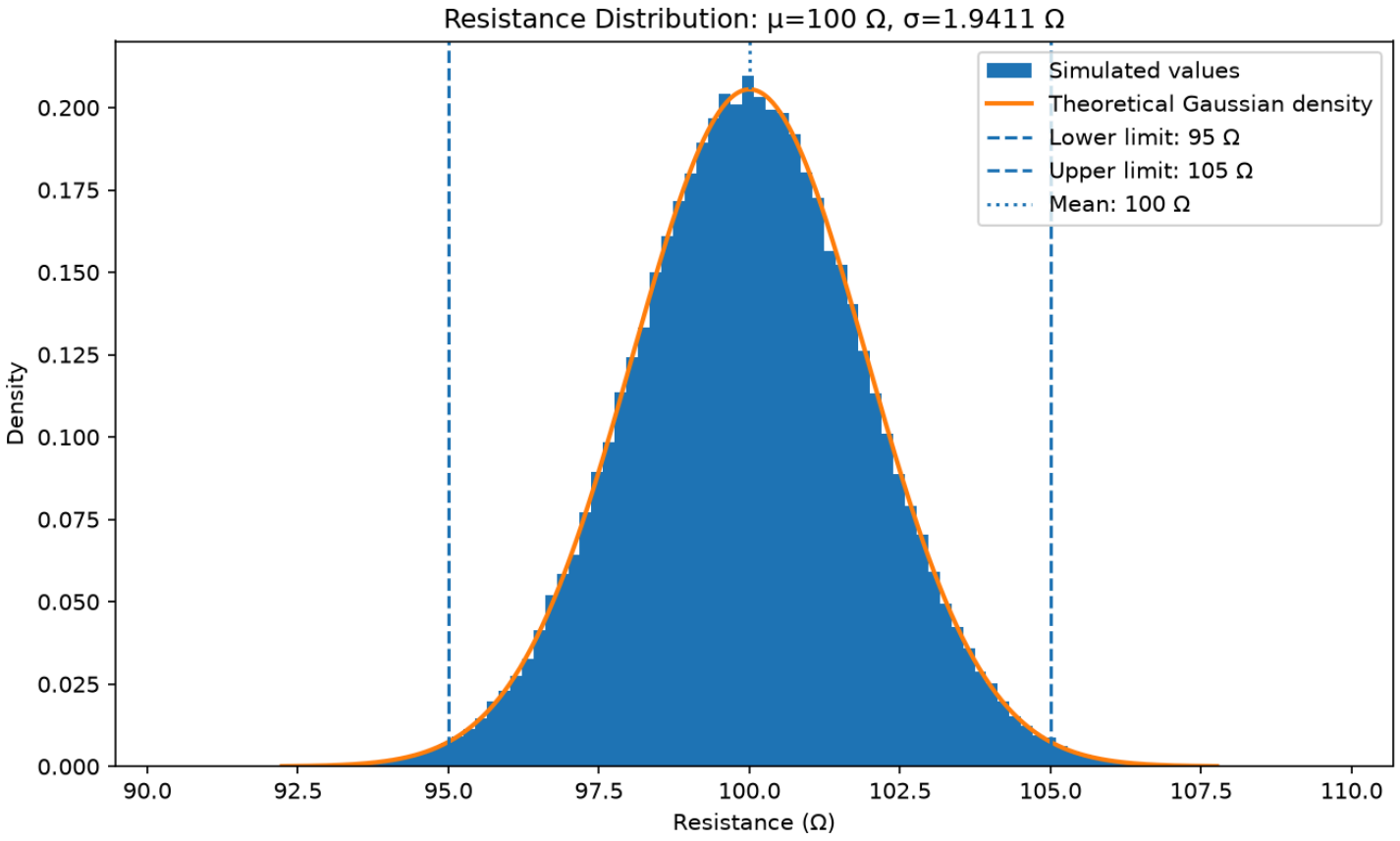
$$2.5758 = \frac{105-100}{\sigma}$$

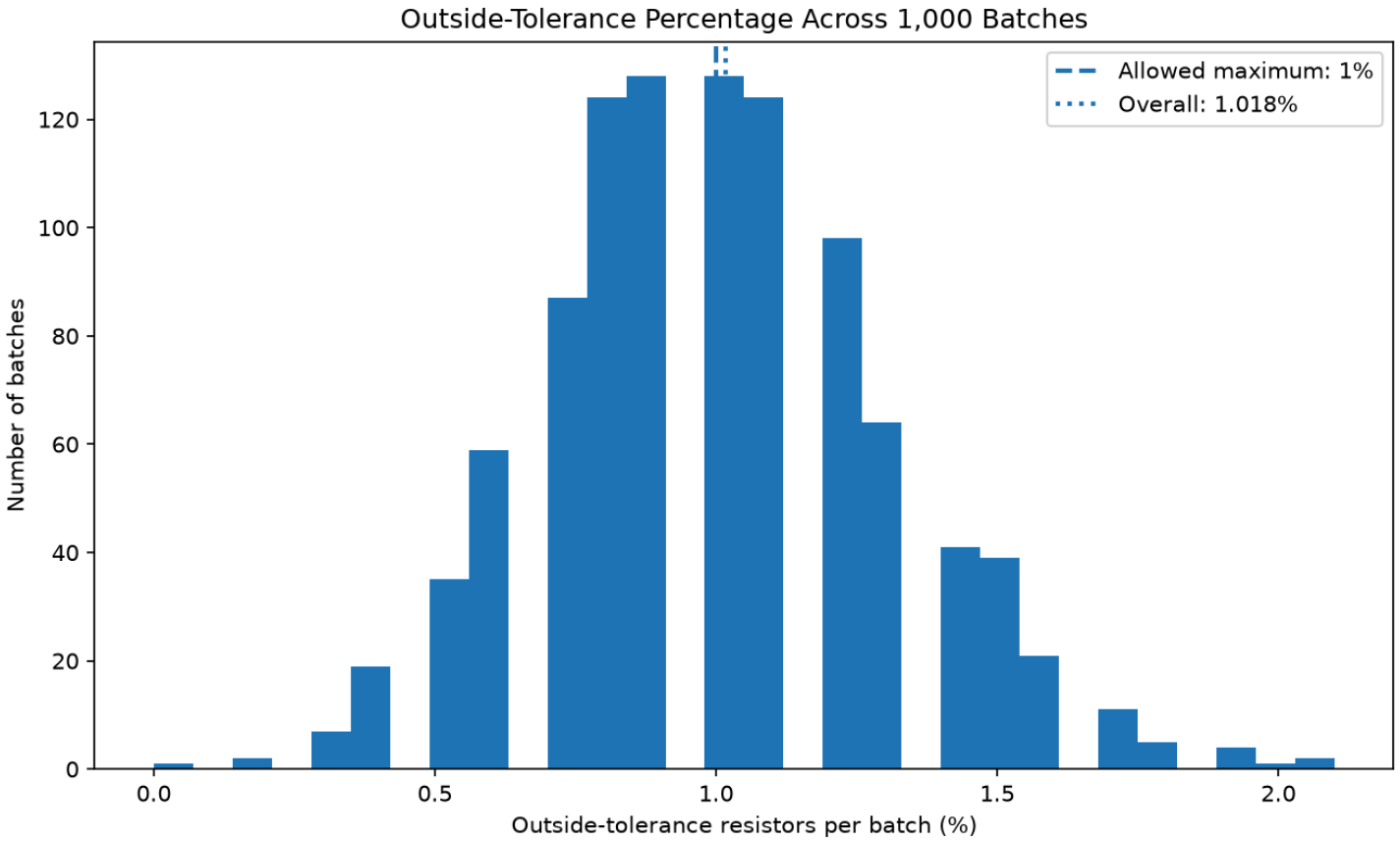
$$\sigma = \frac{5}{2.5758} \approx 1.9411\,\Omega$$

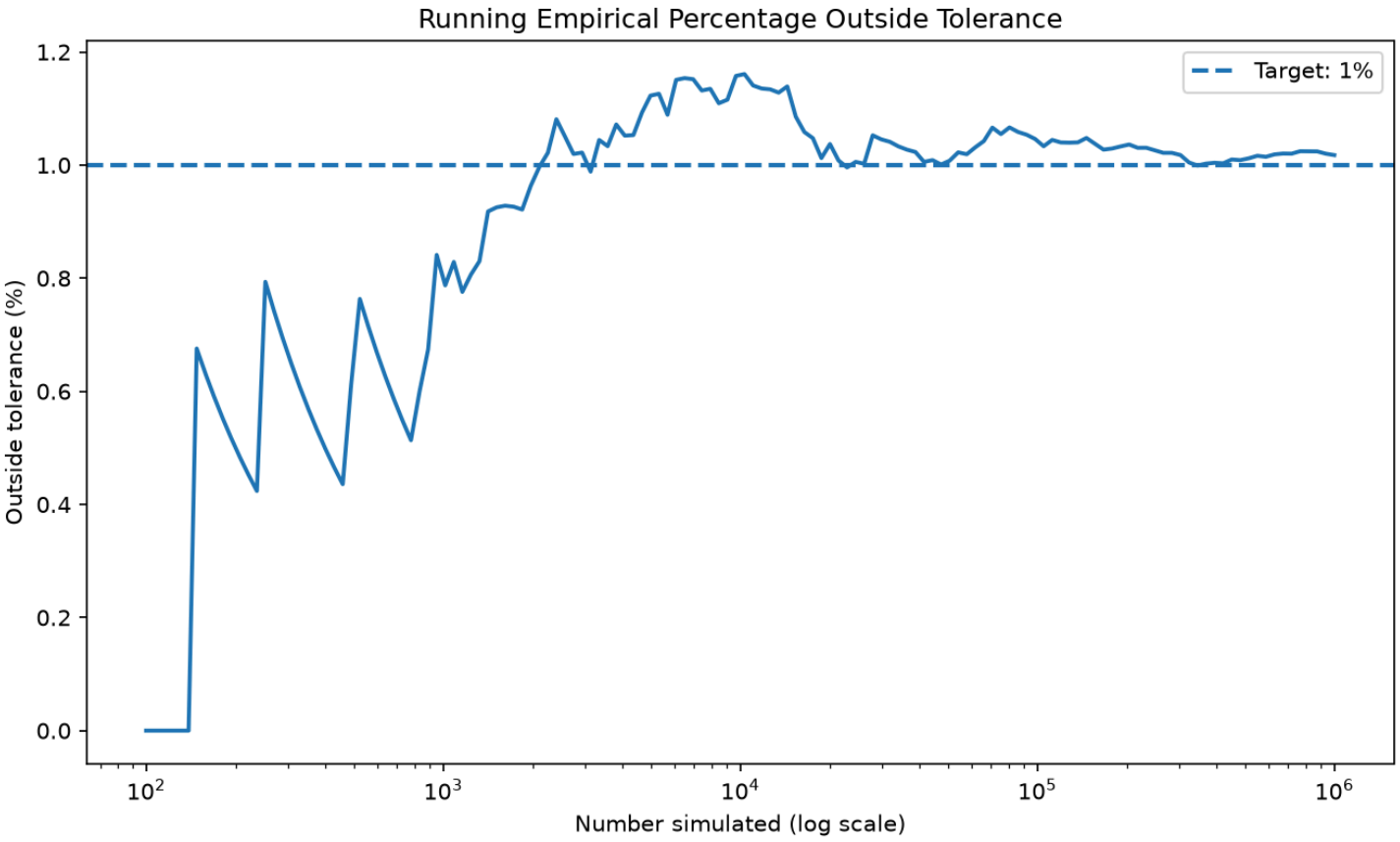


(d) Write a program to verify this, execute the program and generate the plots to empirically show that your choice of σ is correct.

Metric	Value
Maximum sigma (Ω)	1.9411224156473221
Theoretical outside (%)	1.0000000000000009
Simulated resistors	1000000.0
Outside count	10177.0
Empirical outside (%)	1.0177
Empirical within (%)	98.9823
Mean batch outside (%)	0.9945




Outside-Tolerance Percentage Across 1,000 Batches
Allowed maximum: 1%
Overall: 1.018%
Number of batches
0
20
40
60
80
100
120
0.0
0.5
1.0
1.5
2.0
Outside-tolerance resistors per batch (%)


Running Empirical Percentage Outside Tolerance
Target: 1%
Outside tolerance (%)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
10^2
10^3
10^4
10^5
10^6
Number simulated (log scale)

There are 3 plots:

a) Resistance distribution

The histogram shows simulated resistance values. The mean is 100Ω & 99% of the area is between 95 to 105Ω.

b) Defective resistors across batch

Based on CLT, we can see there is a normal distribution. The different batches have defects close to 1% and it is not always above or below it. It averages to 1%.

c) Running empirical defective percentage

The plot shows the percentage of resistors which are defective as the number of observations grow, we can see LLT in action as the value starts to converge to 1%.

Based on the graphs and empirical evidence we can say that the observed defective percentage 1.0177% is very close to 1%. which suggests the chosen σ is correct.